

RESEARCH ARTICLE

Enhancing Thyroid Disease Diagnosis With Machine Learning and Counterfactual Explainable AI

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ABSTRACT Thyroid disease affects millions worldwide and is often underdiagnosed due to complex symptom overlaps and reliance on laboratory interpretation. Machine learning (ML) offers promise for automated diagnosis, but model transparency remains a barrier to clinical adoption. In this study, we develop an interpretable ML framework for thyroid disease classification using a real-world dataset from the UCI repository. We evaluate five classifiers—including Random Forest, CatBoost, and Support Vector Classifier—under a rigorous nested cross-validation scheme to ensure unbiased performance estimation. Our best models achieve near-perfect accuracy (up to 99.7%) and AUC (1.000) on a reduced set of ten consensus physiological features, demonstrating that high performance can be maintained with parsimonious inputs. To enhance trust and clinical utility, we integrate dual explainable AI (XAI) techniques: SHAP for local feature attribution and counterfactual analysis for actionable “what-if” reasoning. These explanations align closely with known endocrine physiology (e.g., TSH and FTI as primary drivers), supporting their face validity. While limited by class imbalance and lack of external validation, our approach combines strong predictive power with transparent decision-making, offering a foundation for trustworthy clinical decision support systems in thyroid care.

INDEX TERMS Thyroid disease diagnosis, machine learning, explainable artificial intelligence, SHAP, counterfactual explanations, clinical decision support.

I. INTRODUCTION

Thyroid is a butterfly-shaped endocrine gland situated anterior to the trachea, just below the larynx. Through secretion of triiodothyronine (T3) and thyroxine (T4) [1], [2], [3], it regulates basal metabolic rate, thermogenesis, cardiovascular tone, and neurocognitive function. Disruption of this axis—whether from autoimmune injury, iatrogenic

causes, or micronutrient imbalance—produces a spectrum of disorders, including hypothyroidism, hyperthyroidism, thyroiditis, goiter, nodules, and malignancy [4], [5], [6]. The clinical burden is substantial: thyroid dysfunction is common, often insidious in onset, and associated with preventable morbidity when diagnosis or treatment is delayed [6], [7], [8], [9], [10]. The gross anatomy of the thyroid gland and an illustration of goiter due to Hashimoto’s thyroiditis are shown in Fig. 1. Standard diagnostic pathways combine

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clinical history and examination with laboratory testing (TSH, free/total T4, T3, thyroid autoantibodies) and selective imaging (ultrasound, radionuclide uptake/scan) [11], [12], [13], [14], [15]. While effective, these workflows face persistent limitations: inter-laboratory variability and thresholding differences, observer dependence for image interpretation, class imbalance in routine cohorts that favors missed minority phenotypes, and latency between specimen collection and decision-making [16], [17], [18]. As a result, real-time, objective decision support that is both accurate and interpretable remains a priority for front-line care. Recent advances in machine learning (ML) have delivered strong performance on thyroid classification tasks spanning structured labs, ultrasound, histopathology, and multimodal data streams [19]. However, three gaps recur. First, many studies restrict to binary or three-way labeling and overlook clinically actionable subclasses. Second, evaluation protocols sometimes conflate model selection and assessment, inflating performance estimates. Third—and most critically for deployment—explanations are often post-hoc, coarse, or non-actionable, limiting clinicians' ability to understand *why* a model predicts a given label and *what* minimal changes could alter it [20], [21].

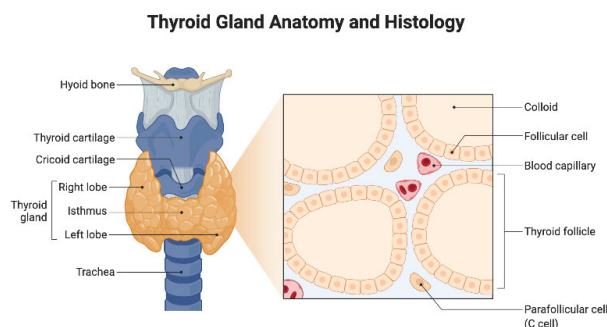


FIGURE 1. Anatomy of the thyroid gland and enlargement (goiter) typical of Hashimoto's thyroiditis. The thyroid lies anterior to the trachea below the larynx; autoimmune inflammation may enlarge the gland and disrupt hormone production [22].

The remainder of this paper is organized as follows. Section II provides a brief clinical background on thyroid disorders and current diagnostic challenges. Section III presents the dataset, feature selection strategies, and machine learning methodology, including nested cross-validation and interpretability components. Section IV reports experimental findings and model comparisons under full and reduced feature sets. Section V discusses the study's limitations and outlines directions for future research. Finally, Section VIII concludes the paper and outlines future research directions.

II. BACKGROUND ON THYROID DISORDERS

A. CLINICAL OVERVIEW OF THYROID DISORDERS

Hyperthyroidism occurs when the thyroid gland produces an excessive amount of T4 hormone, leading to an overactive

thyroid [23], [24]. Common causes include Graves' disease, thyroid nodules, and thyroiditis. Symptoms include tachycardia, unintentional weight loss, arrhythmia, palpitations, and heat intolerance. If left untreated, complications may involve brittle bones, cardiac arrhythmias, and thyrotoxic crisis. Hypothyroidism is an underactive thyroid condition resulting from inadequate hormone production [25], [26], [27]. Etiologies include autoimmune disease, thyroid surgery, radiation therapy, or medication side effects. Symptoms range from fatigue, cold intolerance, and weight gain to depression and memory impairment. Untreated hypothyroidism can progress to severe outcomes such as myxedema or cardiovascular disease. Goiter refers to an abnormal enlargement of the thyroid gland [28], [29], [30]. Causes include iodine deficiency, autoimmune thyroiditis, nodules, or malignancy. Patients may experience neck swelling, difficulty swallowing, or hoarseness.

Thyroiditis denotes inflammation of the thyroid, encompassing Hashimoto's thyroiditis, subacute thyroiditis, and postpartum thyroiditis [31], [32], [33]. Clinical presentation includes neck pain, swelling, and variable hormone release depending on disease stage. Nodules are solid or cystic lumps within the gland [34], [35], [36]. Though often asymptomatic, large nodules can cause compressive symptoms or progress to hyperthyroidism or malignancy. Thyroid cancer arises from malignant transformation of follicular or parafollicular cells [37], [38]. Types include papillary, follicular, medullary, and anaplastic carcinoma. Although early stages may be asymptomatic, progressive disease manifests with neck lumps, voice changes, and regional lymphadenopathy.

B. MACHINE LEARNING APPROACHES

For thyroid disease prediction, machine learning methods have been widely explored across recent studies. Early prediction and stratification into cancer, hyperthyroidism, or hypothyroidism can substantially improve treatment timeliness and outcomes. In [39], a machine learning pipeline was proposed to predict the presence of BRAF mutations in thyroid nodules using ultrasound from 96 cases; Random Forest (RF), Logistic Regression (LR), and Support Vector Machines (SVM) exceeded 60% accuracy. A related effort combined fine-needle aspiration (FNA) cytology with ultrasound features to reduce false negatives in thyroid cancer; RF outperformed gradient methods and decision trees. Building on ultrasound-derived predictors, LR and LASSO were applied in [40] to select malignancy-associated features, followed by RF with a scoring scheme, achieving >80% accuracy. In [41], ten classifiers were benchmarked; Extra Trees yielded 84% accuracy. SVM-based multi-class detection across four thyroid states was reported in [42] with 83.37% accuracy. Decision Trees (DT) were used in [43] to develop a tool (MLTDD) that achieved 98.7% accuracy. Broader molecule-screening pipelines leveraging LR, RF, SVM, Gradient Boosting

TABLE 1. State-of-the-Art Models for Thyroid Disease Prediction.

Ref	Year	Dataset	Model	Results
[59]	2021	UCI new-thyroid dataset	Logistic Regression, Decision Tree, k-Nearest Neighbor	kNN: 96.875% accuracy; DT: 87.5% accuracy; LR: 81.25% accuracy
[60]	2022	UCI thyroid disease dataset	Random Forest (with MLFS feature selection)	RF: 0.99 accuracy, 0.99 F1 score
[61]	2024	Clinical thyroid disease dataset (1232 samples)	Stacking Ensemble (LR, SVM, KNN, DT, ANN)	Stacking Ensemble: 99.9% accuracy, 99.9% sensitivity, 99.8% specificity
[62]	2024	UCI thyroid dataset	AdaBoost (with DE optimization)	AdaBoost: 0.998 accuracy
[63]	2023	UCI thyroid dataset	Random Forest (with LASSO)	RF: 99% accuracy, 97% precision, 92% recall, 95% F1 score
[64]	2024	UCI thyroid dataset	Ensemble ML (Hard Voting on RF and DT)	Ensemble: 99.71% accuracy, 100% sensitivity
[65]	2022	UCI thyroid dataset	Decision Tree, Random Forest	DT: 99.5758% accuracy; RF: 99.3107% accuracy

Machines (GBM), and deep neural networks (DNN) to identify disruptors of thyroid homeostasis were explored in [44]. Feature-engineering comparisons across backward/forward/bidirectional selection and tree-based selection were studied in [45], where Extra Trees—selected features paired with RF reached 99% accuracy and enabled predictions for non-thyroidal syndrome and autoimmune entities. Ensembling and treatment-guidance studies include: meta-classification with boosting/bagging/stacking/voting on a large, complex dataset with careful train–test splits and varied k values in [46]; a treatment-direction predictor for hypothyroid patients using SMOTE and Extra Trees achieving 84% across accuracy/precision/recall/F1 in [43]; and further RF-centered comparative studies reporting 94.8% in [47]. Using the UCI thyroid dataset, six ML algorithms with 10-fold CV were compared in [48]; MLP led on accuracy/recall/F1. A hybrid LDA–KNN–ANFIS pipeline was introduced in [49], combining dimensionality reduction, weighted preprocessing, and neuro-fuzzy inference. A multi-class framework with feature engineering and Extra Trees plus three DL baselines (batch 16, 100 epochs) was reported in [50], evaluated via accuracy, precision, recall, F1, 10-fold CV, SD, and confusion matrices. Additional empirical studies also report strong performance: random forest outperforming peers for hyperthyroidism identification in [45]; multi-model comparisons for three-class thyroidism in [51] with MLP best at 96.4%; and separate analyses distinguishing hyper- vs. hypothyroidism [52] and XGBoost superiority on a UCI dataset [53]. Complementary works explored fuzzy models [54], expert systems [55], swarm optimization for thyroid function analysis [56]. Table 1 provides the comparative analysis of the existing works.

III. METHODOLOGY

A. OUR PROPOSED APPROACH

As illustrated in Figure 2, our proposed pipeline begins with preprocessing (data cleaning and feature engineering), followed by nested cross-validation for model training, tuning, and evaluation, and concludes with post hoc explainability (SHAP) and counterfactual analysis. We treat each supervised learner as a distinct hypothesis family that offers a complementary inductive bias over the same clinical signal. Concretely, we train five models—Support Vector Classifier (SVC), Random Forest (RF), Extremely Randomized Trees (Extra Trees), AdaBoost, and CatBoost—on a single real dataset of thyroid cases with three outcome classes (Normal, Hypothyroid, Hyperthyroid). To probe the trade-off between parsimony and capacity, we evaluate two feature configurations: (i) *FULL* (all available predictors) and (ii) *TOP10* (the consensus set selected via multi-criteria feature selection). Each learner supplies its own “view” of the data: margin-based separation (SVC), bagged axis-aligned partitions (RF), high-randomness ensembles (Extra Trees), stage-wise additive boosting (AdaBoost), and ordered gradient boosting with symmetric trees (CatBoost). This diversity is intentional; it allows us to assess stability of conclusions across fundamentally different decision mechanisms [63], [64].

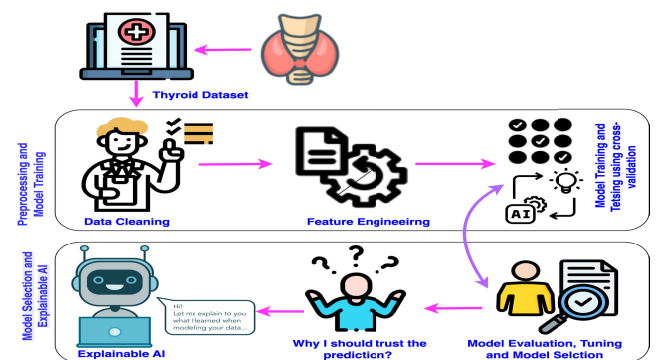


FIGURE 2. Proposed pipeline for thyroid disease prediction: data cleaning, feature engineering, nested cross-validation for model training/selection, and downstream explainability (SHAP) and counterfactual analysis.

All modeling adheres to a strict nested cross-validation protocol (Algorithm 1). In the *outer* loop (stratified 5-fold), we obtain unbiased estimates of generalization. Within each outer training fold, a *stratified inner* loop (3-fold) performs hyperparameter selection via grid search with *accuracy* as the primary score. Model selection is thus confined to the inner loop, while the outer loop is used exclusively for evaluation. This design prevents information leakage from validation to test partitions, and yields robust performance summaries as means and standard deviations across outer folds [65], [66].

B. DATASET DESCRIPTION

For this study, we utilized the **Thyroid Disease dataset** from the UCI Machine Learning Repository, a well-established

Algorithm 1 Nested Cross-Validation for Thyroid Classification (Single Real Dataset)**Require:**

Single dataset $D = \{(x_1, y_1), \dots, (x_N, y_N)\}$
 Feature sets $F \in \{\text{FULL}, \text{TOP10}\}$
 Models $M = \{\text{SVC}, \text{RF}, \text{ExtraTrees}, \text{AdaBoost}, \text{CatBoost}\}$
 Grids $\{G_m\}_{m \in M}$

Ensure:

Outer-CV mean metrics and best hyperparameters
 per model and feature set

1: Preprocessing:

Standardize continuous labs (e.g., TSH, FTI, T3, TT4, T4U) with training-fold statistics; keep binary indicators as $\{0, 1\}$.

No data leakage: fit scalers only on inner/train folds.

2: For each feature set $F \in \{\text{FULL}, \text{TOP10}\}$

3: Select columns X_F according to F ; set labels y (3 classes) 4: Define outer CV as stratified 5-fold with fixed seed

5:for each outer fold $t = 1, \dots, 5$ do

6: Split $(X_F, y) \rightarrow (\text{trainout}, \text{testout})$ stratified by class

7. for each model $m \in M$ do

8: **Model Pipe line:** Build pipeline Π_m

9: **if $m = \text{SVC}$ then**

10: // SVC requires scaling; probability=True for AUC

11: scaler $\rightarrow \text{SVC}(C, \text{kernel}, \text{probability}=\text{True})$

12: **else**

13: // Trees/boosting do not require scaling

14: passthrough $\rightarrow m$

15: **end if**

16: Define inner CV as stratified 3-fold on trainout

17: **Training:** Perform grid search over G_m using inner CV; scoring=accuracy; refit on full trainout to obtain $\hat{\theta}_m^{(t)}$

18: **Evaluation:** Predict on testout: get $\hat{y}_m^{(t)}$ and class probabilities $\hat{P}_m^{(t)}$

19: Compute metrics on testout:

20: Accuracy, weighted Precision/Recall/F1 (per scikit-learn definitions)

21: Multiclass AUC (OVR):

$$\text{AUC}_m^{(t)} = \frac{1}{K} \sum_{c=1}^K \text{AUC}_{-y=c} \left(\hat{P}_m(y = c \mid x) \right)$$

22: Store $\{\text{metrics}_m^{(t)}, \hat{\theta}_m^{(t)}\}$

23: **end for**

24:**end for**

25: **Final Aggregation:** For each model $m \in M$, aggregate over outer folds: report mean $\pm$ std of all metrics; summarize best hyperparameters (e.g., by fold-wise mode or best mean inner-CV score)

benchmark for thyroid disorder classification. This dataset comprises anonymized patient records associated with various thyroid conditions, categorized into three class labels: Normal, Hypothyroid, and Hyperthyroid. The dataset includes a total of 7,200 patient samples, as outlined below:

- **Normal** ($N = 6,666$)
- **Hyperthyroid** ($N = 368$)
- **Hypothyroid** ($N = 166$)

Due to the absence of etiological markers (e.g., TRH or ACTH levels), the dataset does not differentiate between primary and secondary subtypes of hypothyroidism or hyperthyroidism. As a result, this study focuses on high-level

thyroid status classification, deferring subtype differentiation to future work, which will incorporate more granular data, such as pituitary function tests.

The dataset consists of 21 clinical features, which include demographic indicators (e.g., age, sex), treatment history (e.g., on_thyroxine, thyroid_surgery), and laboratory biomarkers (e.g., TSH, T3, TT4, T4U, FTI). These features are commonly used in thyroid function diagnosis. The dataset is notably imbalanced, with most patients labeled as normal or hypothyroid. The features include both *binary clinical attributes* (e.g., on_thyroxine, sick, pregnant) and *continuous biochemical markers* (e.g., TSH, T3, TT4, T4U, FTI).

For data preparation, we merged the two structured text files provided by UCI—ann-train.data and ann-test.data—into a single dataset. The final dataset includes a total of 7,200 samples, each with 21 input features, and a multiclass target variable indicating thyroid disease classification.

The ann-train and ann-test files from the UCI repository were concatenated to form a single dataset of 7,200 samples. Since the dataset does not contain patient identifiers or longitudinal markers, we treated each record as an independent observation, consistent with prior work using this benchmark. Prior to model training, we performed exact-record deduplication by hashing all feature rows and confirmed that no duplicate rows were present. All preprocessing and feature selection steps were nested strictly within cross-validation folds to prevent information leakage.

C. TABLE 1: BASELINE CHARACTERISTICS BY THYROID CLASS

Table 2 presents a summary of baseline patient characteristics stratified by thyroid class. Key findings include:

- **Age** was similar across classes, with normalized means ranging between 0.518 and 0.554.
- **Sex** showed a statistically significant difference ($p < 0.001$), with a higher proportion of females in the hypothyroid group.
- **Biomarkers** such as TSH, T3, TT4, T4U, and FTI displayed expected trends and highly significant differences ($p < 0.001$).
- **Medication-related variables**, such as on_thyroxine and on_antithyroid_medication, also showed significant differences across classes.
- Sparse features like pregnant, lithium, goitre, and tumor were retained for completeness and future explainability analysis.

Continuous variables were scaled to the range of $[0,1]$, and binary variables were encoded as “Yes”/“No” to enhance clarity and interpretability in downstream modeling and XAI visualization.

The demographic and clinical characteristics of three groups (Group 1, $n = 166$; Group 2, $n = 368$; Group 3,

TABLE 2. Baseline characteristics by thyroid class.

Variable	Hypothyroid (N=166)	Hyperthyroid (N=368)	Normal(N=6666)	Overall (N=7200)	P-value
Age (years)					0
Mean (SD)	0.536 (0.191)	0.554 (0.188)	0.518 (0.189)	0.521 (0.189)	
Median [Min, Max]	0.560 [0.0100, 0.890]	0.590 [0.0100, 0.910]	0.540 [0.0100, 0.970]	0.550 [0.0100, 0.970]	
Sex					< 0.001
Female	132 (79.5%)	287 (78.0%)	4590 (68.9%)	5009 (69.6%)	
Male	34 (20.5%)	81 (22.0%)	2076 (31.1%)	2191 (30.4%)	
On thyroxine					< 0.001
No	149 (89.8%)	368 (100%)	5743 (86.2%)	6260 (86.9%)	
Yes	17 (10.2%)	0 (0%)	923 (13.8%)	940 (13.1%)	
Query thyroxine					0.42
No	166 (100%)	363 (98.6%)	6560 (98.4%)	7089 (98.5%)	
Yes	0 (0%)	5 (1.4%)	106 (1.6%)	111 (1.5%)	
On antithyroid medication					0.72
No	165 (99.4%)	365 (99.2%)	6578 (98.7%)	7108 (98.7%)	
Yes	1 (0.6%)	3 (0.8%)	88 (1.3%)	92 (1.3%)	
Sick					0.17
No	165 (99.4%)	352 (95.7%)	6407 (96.1%)	6924 (96.2%)	
Yes	1 (0.6%)	16 (4.3%)	259 (3.9%)	276 (3.8%)	
Pregnant					0.09
No	166 (100%)	368 (100%)	6588 (98.8%)	7122 (98.9%)	
Yes	0 (0%)	0 (0%)	78 (1.2%)	78 (1.1%)	
Thyroid surgery					0.09
No	162 (97.6%)	368 (100%)	6569 (98.5%)	7099 (98.6%)	
Yes	4 (2.4%)	0 (0%)	97 (1.5%)	101 (1.4%)	
I-131 treatment					0.25
No	160 (96.4%)	362 (98.4%)	6557 (98.4%)	7079 (98.3%)	
Yes	6 (3.6%)	6 (1.6%)	109 (1.6%)	121 (1.7%)	
Query hypothyroid					< 0.001
No	142 (85.5%)	317 (86.1%)	6269 (94.0%)	6728 (93.4%)	
Yes	24 (14.5%)	51 (13.9%)	397 (6.0%)	472 (6.6%)	
Query hyperthyroid					0.76
No	158 (95.2%)	344 (93.5%)	6203 (93.1%)	6705 (93.1%)	
Yes	8 (4.8%)	24 (6.5%)	463 (6.9%)	495 (6.9%)	
Lithium					0.49
No	166 (100%)	362 (98.4%)	6581 (98.7%)	7109 (98.7%)	
Yes	0 (0%)	6 (1.6%)	85 (1.3%)	91 (1.3%)	
Goitre					0.17
No	166 (100%)	368 (100%)	6607 (99.1%)	7141 (99.2%)	
Yes	0 (0%)	0 (0%)	59 (0.9%)	59 (0.8%)	
Tumor					0.98
No	162 (97.6%)	360 (97.8%)	6494 (97.4%)	7016 (97.4%)	
Yes	4 (2.4%)	8 (2.2%)	172 (2.6%)	184 (2.6%)	
Hypopituitary					1
No	166 (100%)	368 (100%)	6665 (100.0%)	7199 (100.0%)	
Yes	0 (0%)	0 (0%)	1 (0.0%)	1 (0.0%)	
Psych					0.01
No	165 (99.4%)	358 (97.3%)	6325 (94.9%)	6848 (95.1%)	
Yes	1 (0.6%)	10 (2.7%)	341 (5.1%)	352 (4.9%)	
TSH (mIU/L)					< 0.001
Mean (SD)	0.0963 (0.113)	0.0137 (0.0119)	0.00210 (0.00507)	0.00486 (0.0230)	
Median [Min, Max]	0.0530 [0.00620, 0.530]	0.00970 [0.00610, 0.143]	0.00150 [0, 0.177]	0.00170 [0, 0.530]	
T3					< 0.001
Mean (SD)	0.00965 (0.00592)	0.0173 (0.00543)	0.0204 (0.00746)	0.0200 (0.00754)	
Median [Min, Max]	0.00960 [0.000500, 0.0260]	0.0170 [0.000500, 0.0410]	0.0201 [0.000500, 0.180]	0.0201 [0.000500, 0.180]	
TT4					< 0.001
Mean (SD)	0.0326 (0.0213)	0.0940 (0.0215)	0.112 (0.0345)	0.109 (0.0360)	
Median [Min, Max]	0.0316 [0.00200, 0.0890]	0.0900 [0.0440, 0.150]	0.109 [0.00500, 0.600]	0.107 [0.00200, 0.600]	
T4U					< 0.001
Mean (SD)	0.103 (0.0178)	0.0978 (0.0166)	0.0977 (0.0192)	0.0978 (0.0191)	
Median [Min, Max]	0.103 [0.0610, 0.159]	0.0965 [0.0560, 0.165]	0.0960 [0.0170, 0.233]	0.0970 [0.0170, 0.233]	
FTI					< 0.001
Mean (SD)	0.0323 (0.0204)	0.0969 (0.0196)	0.116 (0.0345)	0.113 (0.0361)	
Median [Min, Max]	0.0325 [0.00200, 0.0640]	0.0950 [0.0649, 0.155]	0.112 [0.00415, 0.642]	0.111 [0.00200, 0.642]	

$n = 6666$) and their combined total ($n = 7200$) were analyzed, with p-values assessing differences across groups. The mean age was similar across groups (0.536 ± 0.191 , 0.554 ± 0.188 , 0.518 ± 0.189 , respectively; overall 0.521 ± 0.189), with medians ranging from 0.540 to 0.590 and no significant difference ($p = 0$). Sex distribution showed a

female predominance (79.5%, 78.0%, 68.9% for Groups 1, 2, and 3), with Group 3 having a higher male proportion (31.1%) compared to Groups 1 (20.5%) and 2 (22.0%; $p < 0.001$). Thyroid-related variables revealed significant differences in thyroxine use (10.2%, 0%, 13.8%; $p < 0.001$), driven by Group 2's absence of thyroxine, and in

querying hypothyroidism (14.5%, 13.9%, 6.0%; $p < 0.001$), with Groups 1 and 2 showing higher rates. No significant differences were found for querying thyroxine ($p = 0.42$), antithyroid medication ($p = 0.72$), hyperthyroidism ($p = 0.76$), thyroid surgery ($p = 0.09$), I-131 treatment ($p = 0.25$), sickness ($p = 0.17$), pregnancy ($p = 0.09$), goitre ($p = 0.17$), or tumors ($p = 0.98$).

Psychiatric conditions were slightly more prevalent in Group 3 (5.1%) than Groups 1 (0.6%) and 2 (2.7%; $p = 0.01$). Thyroid function tests showed significant differences ($p < 0.001$) in TSH (means: 0.0963, 0.0137, 0.00210 mIU/L), T3 (0.00965, 0.0173, 0.0204), TT4 (0.0326, 0.0940, 0.112), T4U (0.103, 0.0978, 0.0977), and FTI (0.0323, 0.0969, 0.116), indicating distinct thyroid profiles, with Group 1 having higher TSH, suggesting possible hypothyroidism, and Group 3 showing higher T3 and TT4, potentially indicating hyperthyroidism.

D. REPRODUCIBILITY AND IMPLEMENTATION DETAILS

To ensure full reproducibility of our results, all experiments were executed with deterministic random seeds. The stratified outer and inner cross-validation splits were generated using a fixed seed (seed = 42) to guarantee identical data partitions across reruns.

Hyperparameter optimization followed predefined search grids for each model, as described in Section II. The same grids were applied consistently in every outer fold, and model selection was performed strictly within the inner loop to avoid information leakage.

All preprocessing operations, including imputation (if applicable) and feature scaling via `StandardScaler`, were fitted exclusively on the training portion of each fold and then applied to validation and test partitions.

Experiments were implemented in Python using `scikit-learn`, `CatBoost`, `NumPy`, and `SHAP`. Because the thyroid dataset originates from the publicly available UCI Machine Learning Repository, independent researchers can replicate the full pipeline directly by following Algorithm 1.

E. EXPLORATORY DATA ANALYSIS

To gain insights into the distribution, correlation, and relationships between variables in the merged thyroid dataset, we conducted a comprehensive exploratory data analysis (EDA), summarized in this section.

1) CLASS DISTRIBUTION

Figure 3 presents the distribution of disease classes within the dataset. The overall distribution highlights the predominance of hyperthyroid (normal thyroid function) cases, followed by a smaller but clinically important set of hypothyroid and hyperthyroid cases. The imbalance observed across classes is typical of real-world clinical data and underscores the importance of synthetic data generation and careful sampling to mitigate model bias. Specifically, the hypothyroid group accounts for a noticeable portion of the dataset and is presumed to include both primary hypothyroidism

(characterized by insufficient thyroid hormone production due to gland dysfunction) and secondary hypothyroidism (typically resulting from pituitary or hypothalamic causes). These subtypes, while related, may differ in severity and clinical presentation, which can influence predictive modeling. However, the dataset does not explicitly distinguish these subtypes due to the absence of etiological markers (e.g., TRH or ACTH levels). In contrast, the hyperthyroid group is comparatively smaller in size, reflecting the lower prevalence of this disorder in the source population. This group is presumed to include subtypes such as primary hyperthyroidism (commonly due to Graves' disease or toxic nodular goiter) and secondary or drug-induced hyperthyroidism. Although less frequent, hyperthyroid cases are clinically significant because of their association with increased cardiovascular risk and metabolic complications. The dataset does not explicitly distinguish subtypes of both hypothyroidism and hyperthyroidism, which are unevenly represented. While primary hypothyroidism is presumed to dominate within the hypothyroid category, hyperthyroid subtypes are fewer but clinically diverse. This distribution provides both opportunities and challenges for classification tasks, especially for models that must capture nuanced patterns across underrepresented categories. We therefore model the three main classes (normal, hypothyroid, hyperthyroid), acknowledging that clinical subtypes exist but are not separable in this cohort, and defer subtype differentiation to future work with more granular data.

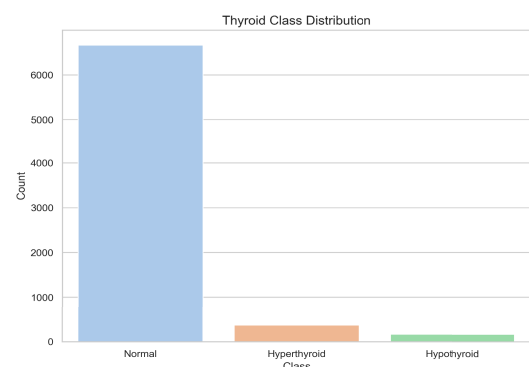


FIGURE 3. Thyroid class distribution: Normal, Hyperthyroid, Hypothyroid.

As illustrated in Figure 3, the dataset exhibits a notable class imbalance, with a majority of samples labeled as Normal, while the Hyperthyroid and Hypothyroid classes are relatively underrepresented. The dataset exhibits moderate class imbalance: Normal (92.6%), Hypothyroid (6.5%), Hyperthyroid (0.9%). While high-performing models achieved strong accuracy, this raises concern about potential bias toward majority classes. This skewed distribution raises the question of whether class balancing techniques—such as Synthetic Minority Over-sampling Technique (SMOTE)—are necessary to mitigate potential model bias toward the majority class. In the current phase of the analysis,

we hypothesize that high-performing classifiers, particularly ensemble models such as Random Forest and CatBoost, may still yield robust results without the need for label balancing, due to their inherent resilience to class imbalance. Thus, we proceed with model training and evaluation on the original (unbalanced) dataset. The rationale is twofold: (1) to assess whether strong predictive performance can be achieved without synthetic sampling, and (2) to preserve the true clinical distribution of thyroid classes, which may be important for real-world deployment. As a point of comparison, we have also conducted an alternative modeling pipeline using SMOTE to balance the class labels. The corresponding results and discussion are provided in Appendix A. The distribution of thyroid disease classes reveals a strong predominance of euthyroid cases compared to hypothyroid and hyperthyroid categories. Within the hypothyroid group, primary hypothyroidism dominates, whereas hyperthyroid cases are fewer but clinically diverse. Detailed visualizations, including class distributions, boxplots of continuous features, categorical variable distributions, and correlation heatmaps, are provided in Appendix A.1 (Figures 11–14) and SMOTE baseline characteristics as Table 12 in Appendix A.2. This dual-track strategy allows us to validate our hypothesis and ensures transparency in reporting model performance across both balanced and unbalanced settings.

2) 3.3.2 CORRELATION AMONG NUMERICAL FEATURES

Figure 4 provides a correlation heatmap of numerical features, revealing strong positive correlations among thyroid-related lab values (e.g., T3, TT4, and FTI with correlations exceeding 0.9). This indicates redundancy in these measurements, which could be addressed through feature selection or dimensionality reduction. Additionally, weak correlations between categorical indicators (e.g., thyroid surgery, I131 treatment) and lab values suggest limited direct influence, warranting further investigation. All continuous variables were standardized using StandardScaler (zero mean, unit variance) within each training fold. No min-max scaling was applied.

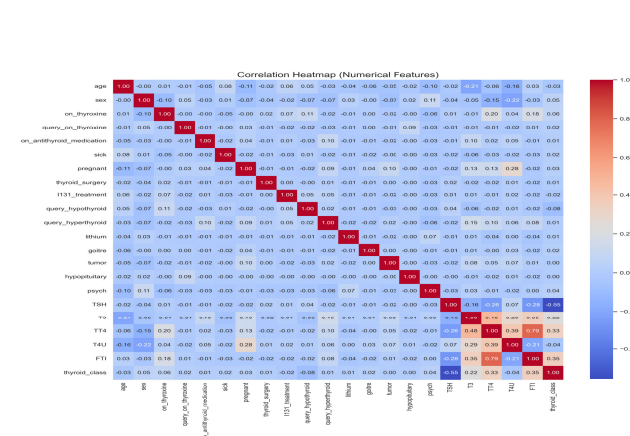


FIGURE 4. Correlation heatmap of numerical variables. Strong correlations were observed among hormone levels.

3) LAB VALUE DISTRIBUTIONS ACROSS CLASSES

Figure 5 presents boxplots of key lab values—TSH, T3, TT4, and T4U—across the three thyroid classes. Notable differences are observed: TSH levels are higher in Hypothyroid cases, T3 and TT4 levels are elevated in Hyperthyroid cases, and T4U shows a broader distribution with higher medians in Hyperthyroid cases. These trends align with clinical expectations, where TSH is typically low in Hyperthyroid and high in Hypothyroid conditions, while T3, TT4, and T4U reflect thyroid hormone activity.

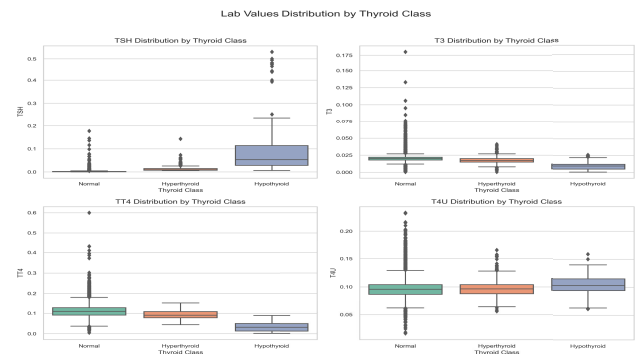


FIGURE 5. Lab values (TSH, T3, TT4, T4U, FTI) stratified by thyroid class.

4) CATEGORICAL VARIABLES BY CLASS

Figure 6 visualizes how categorical clinical indicators vary by thyroid class. Notably, features such as on thyroxine, on antithyroid medication, and query hypothyroid exhibit clear differences in their distribution across the classes. Most categorical features have a majority of 'No' (0) responses, but certain attributes become more prevalent in the diseased groups, aiding interpretability.



FIGURE 6. Distributions of categorical features grouped by thyroid class.

The EDA reveals strong class imbalance and important class-dependent patterns in both numerical and categorical features. Hormonal lab values, in particular, show strong discriminatory power and should be prioritized in modeling efforts. Categorical clinical indicators also provide additional signals. These findings motivate careful feature engineering

and class-balancing strategies for subsequent machine learning tasks.

5) FEATURE SELECTION

To reduce dimensionality, enhance interpretability, and mitigate overfitting in our thyroid classification pipeline, we systematically evaluated a set of filter, wrapper, and embedded feature selection techniques. Given our dataset $D = \{(x_i, y_i)\}_1^N$ with $N = 19,998$ patient instances, where $x_i \in \mathbb{R}^{21}$ is the clinical predictor vector, and $y_i \in \{\text{Normal, Hypothyroid, Hyperthyroid}\}$, we define the goal of feature selection as identifying an optimal subset $S \subseteq \{1, \dots, 21\}$ that maximizes classification performance.

6) MUTUAL INFORMATION (MI)

Mutual information quantifies the reduction in uncertainty of the class label Y given a feature X_j . For a discrete or discretized feature X_j and target Y , the mutual information is defined as [67]:

$$I(X_j; Y) = \sum_{x \in X_j} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (1)$$

In practice, we estimate $p(x, y)$ using empirical frequencies and apply the ‘mutual_info_classif’ function from scikit-learn across all $j = 1, \dots, 21$ features. The top k features with the highest $I(X_j; Y)$ are retained.

7) ANOVA F-STATISTIC (NUMERICAL FEATURES)

For continuous lab variables such as TSH, T3, TT4, and T4U, we employ the one-way ANOVA F-statistic to measure inter-class separability as [68]:

$$F_j = \frac{\sum_{k=1}^K n_k (\bar{x}_{k,j} - \bar{x}_j)^2}{\sum_{k=1}^K \sum_{i \in C_k} (x_{i,j} - \bar{x}_{k,j})^2} \quad (2)$$

where $K = 3$ (number of thyroid classes), n_k is the size of class C_k , $\bar{x}_{k,j}$ is the mean of feature j in class k , and $\bar{x}_j$ is the global mean. This method assumes normally distributed residuals and homoscedasticity.

8) CHI-SQUARED TEST (CATEGORICAL FEATURES)

For binary/categorical indicators such as sex, on thyroxine, tumor, etc., we compute the chi-squared statistic to test independence with thyroid class as [69]:

$$\chi^2 = \sum_r \sum_c \frac{(O_{rc} - E_{rc})^2}{E_{rc}} \quad (3)$$

where O_{rc} is the observed frequency of feature level r and class c , and E_{rc} is the expected frequency under independence. The expected frequency E_{rc} is calculated as:

$$E_{rc} = \frac{(\text{row}_r \text{ total}) \times (\text{col}_c \text{ total})}{N} \quad (4)$$

9) RECURSIVE FEATURE ELIMINATION (WRAPPER-BASED)

Let $f(x; \theta)$ be a base estimator (e.g., Logistic Regression), and let $L(\theta; D)$ be the cross-validated loss. Recursive Feature Elimination (RFE) operates by iteratively removing features that contribute least to performance [70]:

$$S^{(t+1)} = S^{(t)} \setminus \{j^*\} = \arg \min \frac{\partial L}{\partial \theta} \quad (5)$$

We initialize with $S^{(0)} = \{1, \dots, 21\}$ and reduce iteratively until a stopping threshold $|S^{(t)}| = k$ is met.

10) L1-REGULARIZED LOGISTIC REGRESSION (EMBEDDED)

Using a multi-class L1-penalized logistic regression, we induce sparsity in the feature coefficients. The optimization objective is [71]:

$$\min_{\theta} - \sum_{i=1}^N \sum_{c=1}^3 \mathbb{I}(y_i = c) \log \left(\frac{\exp(\theta_c^\top x_i)}{\sum_{c'=1}^3 \exp(\theta_{c'}^\top x_i)} \right) + \lambda \sum_{c=1}^3 \|\theta_c\|_1 \quad (6)$$

Features with zero-valued coefficients $\theta_{cj} = 0$ for all c are removed.

11) RANDOM FOREST FEATURE IMPORTANCE (GINI IMPURITY)

In tree-based models, we compute feature importance based on the average reduction in Gini impurity across all trees and splits [72].

$$\text{Imp}(X_j) = \sum_{s \in \text{splits on } X_j} \Delta_i s \cdot n_s \quad (7)$$

where T is the number of trees, s denotes a split node, $\Delta_i s$ is the impurity decrease at node s , and n_s is the number of samples at node s .

The top 15-ranked features obtained from each of the six methods were compiled and saved for transparency and reproducibility. These full-feature ranking tables are presented in Appendix B, with the following organization:

- Mutual Information (See Table 13)
- ANOVA F-test (See Table 14)
- Chi-Squared Statistic (See Table 15)
- L1-Regularized Logistic Regression (See Table 16)
- Recursive Feature Elimination (RFE) (See Table 17)
- Random Forest Importance (See Table 18)

To enable visual comparison across methods, we additionally present a consolidated grid-based visualization of the top 15 features from each technique in Figure 7. This highlights both consensus and divergence across different algorithmic perspectives on feature relevance, informing subsequent model design decisions.

To identify the most consistently informative features across diverse selection criteria, we adopted a consensus-based strategy. Let $M = \{\text{MI, ANOVA, } \chi^2, \text{L1, RFE, RF}\}$ denote the set of six feature selection methods. For each method $m \in M$, we define the ranked top- k feature set as:

$$F(k) = \{x_j \in X : \text{Rank}_m(x_j) \leq k\} \quad (8)$$

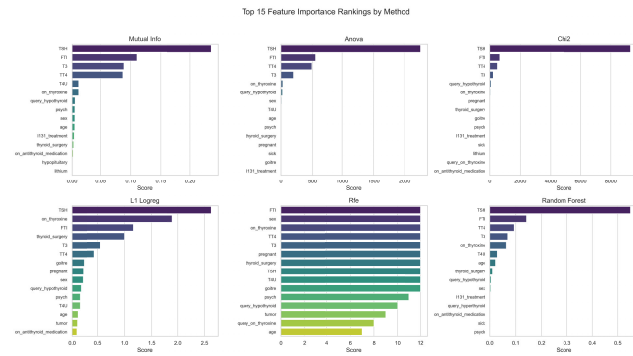


FIGURE 7. Visual comparison of top 15 feature rankings across six feature selection methods: Mutual Information, ANOVA F-test, Chi-Squared, L1-Regularized Logistic Regression, RFE, and Random Forest.

where $\text{Rank}_m(x_j)$ is the importance rank of feature x_j under method m , and X is the full feature space.

We then define the consensus feature subset as the intersection across all methods:

$$F(k)_{\text{consensus}} = \bigcap_{m \in M} F(k)_m \quad (9)$$

In our case, we set $k = 15$ and extracted $F(15)_{\text{consensus}}$, selecting the top $d = 10$ most frequently recurring features across all $F(15)_m$ sets shown in Table 3. This approach emphasizes robustness and model-agnostic agreement in feature utility.

F. MODEL DEVELOPMENT

We selected these five models to represent diverse algorithmic paradigms and inductive biases:

- **SVC:** Margin-based linear separation, sensitive to scaling but robust under high dimensionality.
- **Random Forest & Extra Trees:** Bagged tree ensembles offering stability and native resistance to overfitting.
- **AdaBoost:** Sequential boosting sensitive to weak learners and noise.
- **CatBoost:** Gradient boosting optimized for categorical variables and reduced overfitting via ordered boosting.

This diversity allows us to evaluate performance consistency across fundamentally different decision mechanisms. Tree-based models were prioritized due to their compatibility with exact SHAP computation (TreeSHAP), which ensures higher fidelity explanations compared to approximation methods required for other architectures.

1) PROBLEM SETUP AND NOTATION

Let $D = \{(x_i, y_i)\}_1^n$ denote the training dataset, where $x_i \in \mathbb{R}^{10}$ is the i -th input feature vector derived from clinical and demographic variables, and $y_i \in \{-1, +1\}$ is the corresponding binary label indicating the presence (+1) or absence (−1) of the target thyroid disorder. The feature space is defined as:

$$x = [\text{TSH}, \text{FTI}, \text{T3}, \text{TT4}, \text{on thyroxine},$$

TABLE 3. Top 10 Consensus Features and Their Clinical Interpretations in Thyroid Classification.

Feature	Description and Relevance to Thyroid Function
TSH	Thyroid Stimulating Hormone — Primary regulator of thyroid activity. Elevated TSH typically indicates hypothyroidism; suppressed TSH is seen in hyperthyroidism. Most discriminative marker for thyroid dysfunction.
FTI	Free Thyroxine Index — Adjusted estimate of free T4 levels, accounting for thyroid-binding proteins. Helps differentiate between euthyroid and thyrotoxic states. Useful when total T4 or T3 binding abnormalities are suspected.
T3	Triiodothyronine — Active thyroid hormone. Often elevated in hyperthyroidism. Sensitive in early thyrotoxicosis when T4 may still be normal.
TT4	Total Thyroxine — Total amount of T4 (bound + unbound). Indicates hormone production levels but is influenced by binding proteins. Helps distinguish overt vs. subclinical disease.
on thyroxine	indicates whether the patient is currently taking levothyroxine (synthetic T4). Important for detecting treated hypothyroidism or replacement therapy. Impacts the interpretation of lab values.
hypothyroid flag	Query hypothyroid flag if the patient is suspected to have hypothyroidism but not definitively diagnosed. Can help identify borderline or subclinical cases. Useful for early-stage pattern recognition.
psych	Psychiatric history flag — Thyroid dysfunction, especially hypothyroidism, can present with depressive symptoms, while hyperthyroidism may present with anxiety or agitation. Adds behavioral dimension.
thyroid surgery	indicates a history of thyroidectomy. Strongly affects hormonal regulation and may signal post-operative hypothyroidism or need for hormone replacement.
T4U	Thyroxine Uptake — Proxy for thyroid-binding capacity. Used with TT4 to calculate FTI. Low in hypothyroidism; high in hyperthyroidism. May reflect abnormal protein binding states.
sex	Biological sex — Thyroid diseases are more common in females (esp. autoimmune hypothyroidism). Can moderate clinical presentation and hormone reference ranges. [Ref: NIH Women's Health Initiative].

query hypothyroid, psych, thyroid surgery, T4U, sex]^T where continuous variables are standardized to zero mean and unit variance, and binary indicators (e.g., on thyroxine) are encoded as $\{0, 1\}$, with sex typically mapped as 0 (female) and 1 (male). All features are assumed to lie in a measurable space $X \subseteq \mathbb{R}^{10}$, and the label space is $Y = \{-1, +1\}$.

Our objective is to learn a classifier $f : X \rightarrow Y$ that minimizes the expected risk:

$$R(f) = \mathbb{E}_{(x,y) \sim P} [\ell(f(x), y)], \quad (10)$$

where $\ell : Y \times Y \rightarrow \mathbb{R}_{\geq 0}$ is a loss function, and P is the unknown joint distribution over $X \times Y$. Since P is inaccessible, we minimize the empirical risk:

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i), \quad (11)$$

regularized appropriately depending on the model class.

We employ five state-of-the-art supervised learning algorithms: Support Vector Classifier (SVC), Random Forest (RF), Extra Trees, AdaBoost, and CatBoost. Each is formalized below with full mathematical rigor.

2) SUPPORT VECTOR CLASSIFIER (SVC)

The SVC seeks a maximum-margin hyperplane in a high-dimensional reproducing kernel Hilbert space (RKHS) [73] $\mathcal{H}$ induced by a Mercer kernel $K : X \times X \rightarrow \mathbb{R}$. Let $\varphi : X \rightarrow \mathcal{H}$ be the associated feature map such that $K(x_i, x_j) = \langle \varphi(x_i), \varphi(x_j) \rangle_{\mathcal{H}}$.

The primal optimization problem is:

$$\min_{w \in \mathcal{H}, b \in \mathbb{R}, \xi \in \mathbb{R}^n} \frac{1}{2} \|w\|_{\mathcal{H}}^2 + C \sum_{i=1}^n \xi_i, \quad (12)$$

subject to:

$$y_i (\langle w, \varphi(x_i) \rangle_{\mathcal{H}} + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i = 1, \dots, n,$$

where w is the normal vector to the hyperplane, b the bias, ξ the slack variables, and $C > 0$ the regularization parameter controlling the trade-off between margin maximization and classification error.

By the representer theorem, the solution admits the form:

$$w = \sum_{i=1}^n \alpha_i y_i \varphi(x_i), \quad \alpha_i \in \mathbb{R},$$

leading to the dual problem:

$$\max_{\alpha} \left(\sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j K(x_i, x_j) \right), \quad (13)$$

subject to $0 \leq \alpha_i \leq C$ and $\sum_{i=1}^n \alpha_i y_i = 0$. We employ the radial basis function (RBF) kernel:

$$K(x_i, x_j) = \exp \left(-\gamma \|x_i - x_j\|_2^2 \right), \quad \gamma > 0. \quad (14)$$

The final decision function is:

$$f_{\text{SVC}}(x) = \text{sign} \left(\sum_{i \in S} \alpha_i y_i K(x_i, x) + b \right), \quad (15)$$

where $S = \{i : \alpha_i > 0\}$ is the set of support vectors.

3) RANDOM FOREST (RF)

Random Forest is an ensemble of T decorrelated decision trees $\{h_t(x; \Theta_t)\}_T$, where Θ_t are independent random vectors generated afresh for each tree, inducing randomness in both data sampling and feature selection [74].

Each tree h_t is trained on a bootstrap sample $D_t \subset D$ of size n , and at each internal node, a subset of $m \ll 10$ features is selected uniformly at random from the 10 available features to determine the optimal split. The split criterion maximizes the information gain (or minimizes the Gini impurity) over the selected features.

Let $G(x, j, \tau)$ denote the Gini impurity reduction from splitting on feature j at threshold τ :

$$G(x, j, \tau) = H(D_{\text{parent}}) - \sum_k H(D_k), \quad (16)$$

where $H(D) = 1 - \sum_{c \in \{-1, +1\}} p_c^2$, and p_c is the proportion of class c in the node.

The tree recursively partitions the feature space X into axis-aligned regions $\{R_k\}_T$, assigning to each region a class label based on majority vote:

$$h_t(x) = \arg \max_{c \in \{-1, +1\}} \sum_{i: x_i \in R_k, y_i = c} w_i, \quad (17)$$

where w_i may be uniform or bootstrap-weighted.

The ensemble prediction is:

$$f_{\text{RF}}(x) = \text{sign} \left(\frac{1}{T} \sum_{t=1}^T h_t(x; \Theta_t) \right), \quad (18)$$

with probability estimates given by:

$$P(y = +1|x) = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(h_t(x) = +1). \quad (19)$$

4) EXTRA TREES (EXTREMELY RANDOMIZED TREES)

The Extra Trees ensemble enhances randomness by eliminating bootstrapping and introducing stochasticity in split thresholds [75]. Each tree h_{ext} is trained on the full dataset D , and at each node, a random subset of m' features is selected. For each candidate feature j , s random split points $\{\tau_{j,s'}\}_{s'}$ are drawn uniformly from the empirical range of feature j in the current node.

The split is chosen as:

$$(j^*, \tau^*) = \arg \max_{j \in F_{\text{rand}}, \tau \in \{\tau_{j,s'}\}} G(x, j, \tau), \quad (20)$$

where F_{rand} is the randomly selected feature subset.

5) ADABOOST (ADAPTIVE BOOSTING)

AdaBoost constructs a strong classifier as a weighted linear combination of weak learners (typically decision stumps), iteratively reweighting the training instances to focus on previously misclassified samples [76].

Let $h_t : X \rightarrow \{-1, +1\}$ be the weak learner at iteration t , selected to minimize the weighted error:

$$\epsilon_t = \sum_{i=1}^n w_i^{(t)} \mathbb{I}(h_t(x_i) \neq y_i), \quad (21)$$

where $w_i^{(t)}$ is the weight of instance i at round t . The initial weights are uniform: $w_i^{(1)} = \frac{1}{n}$.

The coefficient α_t for the weak hypothesis is:

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right), \quad (22)$$

assuming $\epsilon_t < \frac{1}{2}$.

The final strong classifier is:

$$f_{\text{AdaBoost}}(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right). \quad (23)$$

6) CATBOOST

CatBoost is a gradient boosting framework that handles categorical features natively and reduces prediction shift via ordered boosting. It constructs an additive model [77]:

$$f_M(x) = f_0 + \sum_{m=1}^M \gamma_m h_m(x), \quad (24)$$

where f_0 is the initial approximation (e.g., log-odds of class priors), h_m are weak learners (symmetric decision trees), and γ_m are coefficients determined by line search.

CatBoost employs an ordered boosting mechanism to mitigate overfitting. It creates a random permutation σ of the training indices and defines an “internal time” $t_i = \sigma(i)$ for each instance. The model at step m is trained on a simulated history, using only instances j with $t_j < t_i$ to compute gradients and construct trees.

The final prediction is:

$$f_{\text{CatBoost}}(x) = \text{sign}(f_M(x)), \quad (25)$$

with probabilities obtained via the logistic link:

$$P(y = +1|x) = \sigma(f_M(x)), \quad (26)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

7) FEATURE SPACE AND MODEL INTEGRATION

All models operate on the same preprocessed feature vector $x \in \mathbb{R}^{10}$, with categorical variables binarized and continuous ones standardized. The joint modeling strategy enables a comprehensive comparison of margin-based, ensemble, and boosting paradigms in diagnosing thyroid dysfunction from clinical markers. Hyperparameters for all models are optimized via nested cross-validation to avoid bias, ensuring statistical validity of performance estimates (See algorithm 1).

G. INTERPRETABILITY: SHAP AND COUNTERFACTUAL EXPLANATIONS

1) MOTIVATION

Clinical decision support benefits from explanations at two levels: global—which predictors the model relies on across the cohort—and local—why a specific patient was classified as Normal, Hypothyroid, or Hyperthyroid. We therefore pair our classifiers with SHAP (additive feature attributions) and with counterfactual explanations that probe decision boundaries by asking what minimal, plausible change would alter a prediction.

2) GLOBAL AND LOCAL SHAP

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^K$ denote a trained model mapping a $d = 10$ -dimensional input vector $\mathbf{x} = [\text{TSH}, \text{FTI}, \dots, \text{sex}]^\top$ to $K = 3$ class logits (or probabilities). SHAP (SHapley Additive exPlanations) provides a unified framework for local interpretability by assigning to each feature $j \in \{1, \dots, d\}$ a value $\phi_j(\mathbf{x})$ that quantifies its contribution to the deviation of the model output from the average prediction.

Formally, SHAP decomposes the model output into a linear additive explanation model:

$$f_k(\mathbf{x}) = \phi_0^{(k)} + \sum_{j=1}^d \phi_j^{(k)}(\mathbf{x}), \quad \forall k \in \{1, \dots, K\}, \quad (27)$$

where $f_k(\mathbf{x})$ is the output (logit or probability) for class k , $\phi_0^{(k)} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[f_k(\mathbf{x})]$ is the global expected value (baseline), and $\phi_j^{(k)}(\mathbf{x})$ is the *Shapley value* for feature j in explaining the deviation $f_k(\mathbf{x}) - \phi_0^{(k)}$.

The Shapley value is

$$\phi_j^{(k)}(\mathbf{x}) = \sum_{S \subseteq \{1, \dots, d\} \setminus \{j\}} \frac{|S|!(d-|S|-1)!}{d!} [f_k(\mathbf{x}_{S \cup \{j\}}) - f_k(\mathbf{x}_S)], \quad (28)$$

with $f_k(\mathbf{x}_S)$ the conditional expectation

$$f_k(\mathbf{x}_S) = \mathbb{E}_{\mathbf{x}_{\bar{S}} \sim \mathcal{D}_{\bar{S}|S}} [f_k(\mathbf{x}_S, \mathbf{x}_{\bar{S}})], \quad (29)$$

where $\bar{S}$ is the complement of S .

Exact computation is exponential in d . For tree-based models (RF, Extra Trees, AdaBoost, CatBoost) we use **TreeSHAP** [78]. For SVC we use **KernelSHAP** [79], which solves

$$\min_{\phi^{(k)} \in \mathbb{R}^{d+1}} \sum_{S \subseteq \{1, \dots, d\}} \pi(S) \left(f_k(\mathbf{x}_S) - \sum_{j \in S} \phi_j^{(k)}(\mathbf{x}) - \phi_0^{(k)} \right)^2, \quad (30)$$

with kernel $\pi(S) = \frac{d}{\binom{d}{|S|} |S| (d-|S|)}$ emphasizing mid-sized subsets. We report global importance via mean absolute SHAP on held-out data

$$\bar{\Phi}_j^{(k)} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} |\phi_j^{(k)}(\mathbf{x}_i)|, \quad (31)$$

and local plots for exemplar patients.

3) COUNTERFACTUAL EXPLANATIONS

Given an input x with predicted class $\hat{y} = \arg \max_k f_k(x)$ and target class $y^* \neq \hat{y}$, we seek a counterfactual x^* that minimally perturbs x while flipping the decision:

$$x^* \in \arg \min_{x' \in F} (\|W(x' - x)\|_1 + \lambda \ell(f(x'), y^*)), \quad (32)$$

subject to the following constraints:

$$x'_j = x_j \quad \forall j \in \mathcal{X}, \quad (33)$$

$$l \leq x' \leq u, \quad (34)$$

where:

- $\|W(x' - x)\|_1$ is a weighted sparsity cost, with $W = \text{diag}(w_1, \dots, w_d)$ and $w_j = \frac{1}{\text{MAD}_j}$ to normalize units.
- $\ell(f(x'), y^*)$ is a loss function encouraging class y^* , e.g., $\ell = -\log P(y^*|x') + \max_{k \neq y^*} \log P(k|x')$.
- $\mathcal{X}$ indexes immutable features (e.g., sex, thyroid surgery); only labs {TSH, FTI, T3, TT4, T4U} vary.

- l, u are bounds from empirical percentiles (e.g., 1st–99th) to ensure plausibility.

We solve (43) via projected coordinate descent guided by local SHAP magnitudes, updating a mutable feature $j \notin \mathcal{X}$ by:

$$x' \leftarrow \Pi_{[l,u]}(x' - \eta \nabla_{x'} (\|W(x' - x)\|_1 + \lambda \ell(f(x'), y^*))), \quad (35)$$

until the prediction flips to y^* or a maximum number of iterations is reached.

We report:

- The scaled distance $\Delta = \|W(x^* - x)\|_1$ from (43);
- Per-feature deltas $\delta_j = x^* - x_j$ for all mutable $j \notin \mathcal{X}$;
- The decision shift $\Delta f_k = f_k(x^*) - f_k(x)$ for each class k .

These quantify how small, clinically bounded edits to thyroid labs can alter model diagnoses.

H. PERFORMANCE AND EVALUATION METRICS

1) THREE-CLASS FORMULATION

Let $D = \{(x_i, y_i)\}_N$ denote the test set, with $x_i \in \mathbb{R}^d$ and $y_i \in \{1, 2, 3\}$ corresponding to Normal, Hypothyroid, and Hyperthyroid, respectively. Define the one-hot label matrix $Y \in \{0, 1\}^{N \times 3}$, where $Y_{ik} = \mathbb{1}\{y_i = k\}$, and the model's class-probability matrix $P \in [0, 1]^{N \times 3}$, where $P_{ik} = P(y = k | x_i)$.

The predicted labels are $\hat{y}_i = \arg \max_k P_{ik}$, and the cross-entropy on the test set is:

$$L_{CE} = - \sum_{i=1}^N \sum_{k=1}^3 Y_{ik} \log P_{ik}. \quad (36)$$

2) CONFUSION MATRIX AND DERIVED METRICS

Let the confusion matrix $C \in \mathbb{N}^{3 \times 3}$ be defined with rows corresponding to true classes and columns corresponding to predicted classes:

$$C = \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}$$

Overall accuracy is the normalized trace:

$$\text{Acc} = \frac{\text{tr}(C)}{N} = \frac{C_{11} + C_{22} + C_{33}}{N}. \quad (37)$$

For class $k \in \{1, 2, 3\}$, define the following metrics:

$$TP_k = C_{kk}, \quad FN_k = \sum_{j \neq k} C_{kj}, \quad FP_k = \sum_{i \neq k} C_{ik},$$

$$TN_k = \sum_{i,j \neq k} C_{ij}.$$

Then:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \quad (38)$$

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \quad (39)$$

$$F1_k = 2 \cdot \frac{\text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \quad (40)$$

Let the per-class supports be $s_k = \sum_{i=1}^3 C_{kj}$ and $S = \sum_{k=1}^3 s_k = N$. We report macro and support-weighted averages:

$$\text{Precision}_{\text{macro}} = \frac{1}{3} \sum_{k=1}^3 \text{Precision}_k,$$

$$\text{Precision}_{\text{weighted}} = \frac{1}{N} \sum_{k=1}^3 s_k \cdot \text{Precision}_k,$$

$$\text{Recall}_{\text{macro}} = \frac{1}{3} \sum_{k=1}^3 \text{Recall}_k,$$

$$\text{Recall}_{\text{weighted}} = \frac{1}{N} \sum_{k=1}^3 s_k \cdot \text{Recall}_k,$$

$$F1_{\text{macro}} = \frac{1}{3} \sum_{k=1}^3 F1_k, \quad F1_{\text{weighted}} = \frac{1}{N} \sum_{k=1}^3 s_k \cdot F1_k. \quad (41)$$

I. OVR ROC-AUC FROM THE PROBABILITY MATRIX

For one-vs-rest (OvR) ROC per class k , regard class k as positive and $\{1, 2, 3\} \setminus \{k\}$ as negative, using scores $\{P_{ik}\}_N$. The ROC curve is:

$$TPR(t) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}\{y_i = k\} \mathbb{1}\{P_{ik} \geq t\},$$

$$FPR(t) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}\{y_i \neq k\} \mathbb{1}\{P_{ik} \geq t\}.$$

The per-class AUC is:

$$\text{AUC}_k = \int_0^1 TPR_k^{-1}(u) \cdot FPR_k(u) du. \quad (42)$$

We summarize with:

$$\text{AUC}_{\text{macro}} = \frac{1}{3} \sum_{k=1}^3 \text{AUC}_k,$$

$$\text{AUC}_{\text{weighted}} = \frac{1}{N} \sum_{k=1}^3 s_k \cdot \text{AUC}_k.$$

IV. RESULTS

We evaluate thyroid status classification (Normal, Hypothyroid, Hyperthyroid) using the real UCI dataset (Section 3.2), our top-10 consensus features (TSH, FTI, T3, TT4, T4U, on thyroxine, query hypothyroid, psych, thyroid surgery, sex), and the nested cross-validation (CV) protocol in Algorithm 1. Unless otherwise stated, all summary metrics, confusion matrices, and ROC curves are computed from out-of-fold (OOF) predictions from the 5-fold outer CV, thereby reflecting performance strictly on unseen data within each fold.

Standardization was applied only to continuous laboratory variables (TSH, FTI, T3, TT4, T4U) within the inner training fold using StandardScaler. The fitted scaler was then used to

transform validation and test splits, preventing data leakage. Binary/categorical features (e.g., sex, on thyroxine) were kept as {0,1}. Scaling was disabled for tree-based models (RF, CatBoost, etc.) since they do not require it; only SVC pipelines included scaling as a mandatory step.

Table 4 summarizes the nested cross-validation (CV) results for five classifiers under two feature regimes: the full feature set and the top-10 consensus features. Hyperparameters were optimized using grid search over predefined ranges with a 3-fold inner CV, targeting accuracy. Final models were retrained on the full outer training folds with the best parameters identified.

Across all models, accuracy exceeded 0.96 and AUC was 0.990, indicating a relatively separable three-class classification task. Ensemble methods performed particularly well: Random Forest and CatBoost consistently achieved the best mean scores across both feature regimes (Accuracy/F1 0.996–0.997; AUC 0.999–1.000), while SVC trailed slightly (Accuracy 0.965–0.967; AUC 0.990–0.992). Performance with the top-10 features was either preserved or marginally improved compared to the full feature set (e.g., Extra Trees: 0.981 $\rightarrow$ 0.989 accuracy; Random Forest: 0.996 $\rightarrow$ 0.997 accuracy), consistent with the hypothesis that a small set of physiologically salient labs and flags carry most of the predictive signal.

Searched hyperparameter ranges and best configurations:

- SVC was tuned over $C \in \{0.1, 1, 10, 100\}$, kernels {linear, rbf}, and $\gamma \in \{\text{scale}, \text{auto}\}$. The best-performing model used $C = 1$, kernel=linear, $\gamma = \text{scale}$.
- Random Forest was optimized over $n_{\text{estimators}} \in \{50, 100, 200\}$, $\text{max_depth} \in \{\text{None}, 10, 20\}$, and criterion {gini, entropy}. The best setting was $n_{\text{estimators}} = 100$, $\text{max_depth}=\text{None}$, criterion=gini.
- Extra Trees was searched over identical ranges to RF, with $n_{\text{estimators}} = 200$, $\text{max_depth}=\text{None}$, criterion=gini selected as optimal.
- AdaBoost was tuned with $n_{\text{estimators}} \in \{50, 100, 200\}$ and $\text{learning_rate} \in \{0.5, 1.0, 1.5\}$, converging on $n_{\text{estimators}} = 50$, $\text{learning_rate}=1.0$.
- CatBoost was optimized over depth {4, 6, 8}, iterations {100, 200}, $\text{learning_rate} \in \{0.03, 0.1, 0.3\}$, and $l2_{\text{leafreg}} \in \{1, 3, 5\}$. The best configuration was depth=4, iterations=200, $\text{learning_rate}=0.1$, $l2_{\text{leafreg}} = 3$.

Given this profile, Random Forest was designated as the primary model for reporting and deployment: it delivers state-of-the-art performance, produces well-calibrated probabilities, requires minimal preprocessing, and supports exact TreeSHAP explanations. CatBoost is retained as a near-identical, interpretable baseline.

The outstanding performance of Random Forest and CatBoost arises from their ability to capture nonlinear interactions among laboratory values while avoiding overfitting through ensemble averaging. Their hierarchical splitting structure aligns with clinical reasoning (e.g., “if TSH is

elevated $\rightarrow$ check FTI”). By contrast, SVC, although competitive, is less effective in multi-class settings without kernel fine-tuning.

Finally, to examine the impact of class imbalance, all models were retrained using SMOTE-balanced data. As detailed in Appendix C (Table 18), minority-class recall improved modestly (<2% for hyperthyroid cases), while precision declined slightly due to increased false positives. Therefore, we retained the original imbalanced setup to reflect real-world prevalence and avoid synthetic artifacts.

V. DISCUSSION

This study presents an interpretable machine learning framework for three-class thyroid status classification using a publicly available clinical benchmark dataset. By combining rigorous nested cross-validation, consensus-based feature reduction, SHAP attribution, and counterfactual reasoning, we aimed to balance predictive performance with transparency required for clinical decision support.

A. PRINCIPAL FINDINGS

Across all evaluated learners, ensemble methods—particularly Random Forest and CatBoost—achieved extremely strong discrimination, with AUC values approaching unity and accuracies above 99% in several configurations. Importantly, performance was preserved when reducing the input space from the full feature set to the Top-10 consensus variables. This supports the hypothesis that a compact group of physiologically meaningful markers (TSH, FTI, T3, TT4, T4U) together with therapy and demographic indicators carries the dominant predictive signal for thyroid dysfunction.

The ability to maintain performance under aggressive feature reduction enhances interpretability, simplifies deployment, and reduces dependency on rarely available inputs.

B. WHY ARE THE METRICS SO HIGH?

Near-perfect AUC values in a clinical ML study warrant careful interpretation. Several factors likely contribute.

First, the UCI dataset is relatively clean and exhibits strong biochemical separation between classes, particularly via TSH and FTI. Second, the learning problem is supervised with well-defined labels, and the hormonal markers are mechanistically tied to disease states. Third, ensemble methods are highly effective at capturing nonlinear threshold behavior common in endocrine diagnostics.

To guard against overly optimistic estimates, we employed strict nested cross-validation, confined model selection to inner folds, and evaluated performance exclusively using out-of-fold predictions. We also verified that no duplicate feature rows were present after merging the original files. Nevertheless, such performance should be interpreted as evidence of separability within this benchmark rather than guaranteed real-world clinical accuracy.

TABLE 4. Nested CV on the Real Thyroid Dataset. Metrics are outer-fold means (3 d.p.).

Model	Feat.	Acc.	Rec.	Prec.	F1	AUC	Best params
SVC	Full	0.965	0.965	0.962	0.962	0.992	C=1, kernel=linear, gamma=scale
SVC	Top-10	0.967	0.967	0.965	0.963	0.990	
Random Forest	Full	0.996	0.996	0.996	0.996	0.999	n_estimators=100, max_depth=None, criterion=gini
Random Forest	Top-10	0.997	0.997	0.997	0.997	1.000	
Extra Trees	Full	0.981	0.981	0.981	0.980	0.998	n_estimators=200, max_depth=None, criterion=gini
Extra Trees	Top-10	0.989	0.989	0.989	0.989	0.999	
AdaBoost	Full	0.990	0.990	0.991	0.990	0.996	n_estimators=50, learning_rate=1.0
AdaBoost	Top-10	0.990	0.990	0.991	0.990	0.996	
CatBoost	Full	0.996	0.996	0.996	0.996	1.000	depth=4, iterations=200, learning_rate=0.1, l2_leaf_reg=3
CatBoost	Top-10	0.996	0.996	0.996	0.996	1.000	

C. IMPACT OF CLASS IMBALANCE

The dataset is dominated by normal cases, with substantially fewer hypothyroid and hyperthyroid samples. While global accuracy and AUC remain high, imbalanced settings can mask reduced sensitivity for minority classes. For this reason, we reported class-wise precision, recall, F1, and confusion matrices, and examined SMOTE-based retraining in supplementary analyses.

We observed modest gains in minority recall at the expense of increased false positives. Because synthetic oversampling may distort the natural prevalence encountered in practice, we retained the original distribution for primary reporting while transparently presenting both views.

D. INTERPRETABILITY AND CLINICAL MEANING

SHAP analysis consistently identified TSH and FTI as dominant drivers, which aligns with established endocrine physiology and provides face validity. However, attribution in the presence of highly correlated laboratory measures must be interpreted cautiously, as importance may be distributed across related variables.

Counterfactual explanations should be understood as model-based sensitivity probes. They illustrate which minimal feature perturbations would alter a prediction according to the learned decision boundary. They are not treatment recommendations and must always be evaluated within clinical feasibility, patient safety, and temporal constraints.

E. REPRODUCIBILITY AND TRANSPARENCY

To promote reproducibility, all preprocessing, feature selection, and hyperparameter optimization were embedded within deterministic nested cross-validation using fixed random seeds. The dataset is publicly available from the UCI repository, enabling independent verification of results. By reporting explicit grids, pipelines, and evaluation protocols, we aim to minimize ambiguity in replication.

VI. WHY THE TOP-10 FEATURE SET FOR THE UI AND DEPLOYMENT?

Although the full feature set yields excellent scores, the Top-10 consensus set (TSH, FTI, T3, TT4, T4U, on thyroxine, query hypothyroid, psych, thyroid surgery, sex) offers multiple advantages with no loss of accuracy:

- **Parsimony without performance loss:** As Table 4 shows, the Top-10 matches or slightly improves Accuracy/F1/AUC for the best models, suggesting that remaining variables are either redundant or weakly informative.
- **Clinically aligned signal:** The selected labs (TSH, FT4/FTI, TT4, T3, T4U) directly reflect hypothalamic pituitary-thyroid axis physiology; therapy flags (on thyroxine, thyroid surgery) encode iatrogenic states; sex is a known modifier. This improves face validity and trust.
- **Interpretability and XAI:** Fewer inputs yield crisper SHAP summaries and more actionable counterfactuals (Sec. 3.6), simplifying communication of “why” and “what-if” to clinicians.
- **Operational robustness:** The Top-10 fields are routinely available in EHRs, reducing missing-data risk, user burden, and latency at prediction time.
- **Regularization by design:** Restricting to the most stable features mitigates spurious correlations and potential shift from rare indicators; this complements statistical regularization used during training.

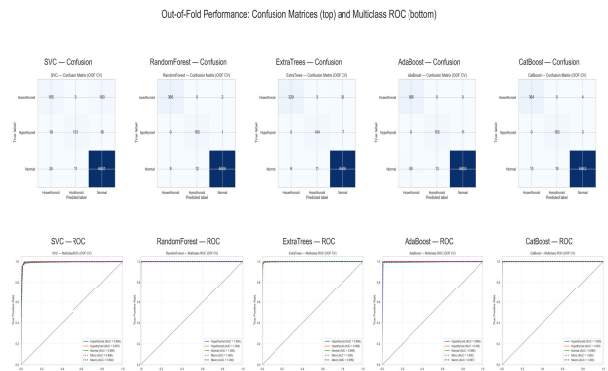


FIGURE 8. Out-of-fold (5-fold outer CV) confusion matrices and ROC curves for all models (SVC, Random Forest, Extra Trees, AdaBoost, CatBoost) using the real dataset and the chosen feature set. Rows show true classes and columns show predicted classes; values are aggregated over outer folds and Curves include per-class ROC along with micro- and macro-average ROC; AUCs are computed from OOF probabilities and aggregated across folds.

Figure 8 and 9 present, for each candidate model, the out-of-fold (OOF) confusion matrix (top row) alongside

TABLE 5. Mathematical structure and clinical interpretation of SHAP components.

Equ	Mathematical Meaning	Clinical Interpretation
(1)	$f_k(x) = \varphi_k + \sum_{j=1}^d \varphi_j(x)$	Decomposes prediction into baseline and feature effects. Patient's predicted risk = population baseline + personalized adjustments from biomarkers.
(2)	$\varphi_0 = \mathbb{E}_{x \sim D}[f_k(x)]$	Expected model output for class k . Represents average risk of class k in the reference population.
(3)	$\varphi_j(x) = \sum_{S \subseteq N \setminus \{j\}} \frac{ S !(d- S -1)!}{d!} [f_k(x_S \cup \{j\}) - f_k(x_S)]$	Shapley value for marginal contribution of feature j . Quantifies how much each clinical variable contributes to diagnosis for class k .
(4)	$f_k(x_S) = \mathbb{E}_{\bar{x} \sim D}[f_k(x_S, \bar{x})]$	Conditional expectation over missing features. Model's estimated output when only some features are known (e.g., partial labs).
(5)	$\pi(S) = \frac{d!}{ S !(d- S)!}$	Kernel weight for subset S in KernelSHAP. Prioritizes balanced subsets to stabilize approximation.
(6)	$\Phi(k) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \varphi_j(x_i) $	Average absolute SHAP value. Ranks, which features the most influential predictions for class k across all patients.

the corresponding multiclass ROC curves (bottom row). Across all tree and boosting methods—Random Forest, ExtraTrees, AdaBoost, and CatBoost—the ROC curves hug the upper-left corner, yielding micro and macro AUCs essentially at or extremely close to 1.000. The linear SVC performs strongly but consistently below the ensembles, with micro/macro AUCs around 0.994–0.996, indicating a mild loss of discrimination when class boundaries are non-linear. The confusion matrices corroborate these findings: most predictions lie on the diagonal, especially for the Normal class, while the few misclassifications that remain are predominantly Hyperthyroid (and to a lesser extent Hypothyroid) cases labeled as Normal, pointing to a tight decision boundary between these two phenotypes.

Among the ensembles, CatBoost and Random Forest exhibit the cleanest matrices, with vanishingly small off-diagonal counts and ROC curves that are effectively saturated. ExtraTrees achieves a similar AUC profile but shows slightly more spillover from Hyper/Hypo into Normal. AdaBoost also attains near-perfect AUCs, though its confusion matrix reveals a few additional NormalX→Hyper, Hypo assignments, suggesting a somewhat sharper boundary that occasionally over-flags positives. SVC's matrix shows the highest rate of Hyper→Normal errors among the models, which is consistent with the limitations of a linear separator in the presence of feature interactions.

Taken together, these results indicate that non-linear relations among the top clinical variables (notably TSH, FTI, T3, TT4, T4U, and key indicators such as on thyroxine and query hypothyroid) are critical for accurate triage across Normal, Hypothyroid, and Hyperthyroid. For deployment, CatBoost or Random Forest provide the best balance of sensitivity and specificity with minimal cross-class leakage, whereas SVC remains a useful linear baseline but risks missing Hyperthyroid presentations. In settings where the clinical cost of false negatives is asymmetric, threshold calibration on the probabilistic outputs of the tree/boosting models can further bias decisions toward safeguarding recall

in the disordered classes without materially sacrificing overall performance.

Deep learning models were excluded due to the moderate size of the UCI thyroid dataset (7,200 samples), which increases the risk of overfitting without large-scale pretraining or extensive regularization. Moreover, our focus was on fully interpretable pipelines, where SHAP values can be computed exactly using TreeSHAP for tree ensembles. Most deep networks require KernelSHAP or similar approximations, which are computationally expensive and less stable. Future work may explore lightweight neural architectures with integrated explainability or hybrid symbolic-AI frameworks. Including deep models would also complicate counterfactual generation due to non-convex loss landscapes and longer inference times—factors critical for clinical deployment feasibility.

A. INTERPRETABILITY

1) EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI):

MATHEMATICAL FOUNDATION AND CLINICAL VALIDATION

To ensure transparency and clinical fidelity, we ground our interpretability analysis in the axiomatic framework of SHAP (SHapley Additive exPlanations), which uniquely satisfies desirable properties such as local accuracy, missingness, and consistency [80]. Below, we present a formal derivation of the SHAP decomposition, validate its mathematical soundness, and provide clinical interpretations through structured analysis of representative patient cases.

This table bridges abstract mathematics and clinical reasoning. For example, if $\varphi_j(x) > 0$, it indicates that feature j pushes the model toward class k , while $\varphi_j(x) < 0$ suggests it detracts from it. The additive structure ensures no interaction terms are hidden, making the model fully decomposable.

The global SHAP analysis confirms that:

- **TSH** is the most influential feature by a wide margin, reflecting its central role in thyroid physiology and clinical diagnosis.

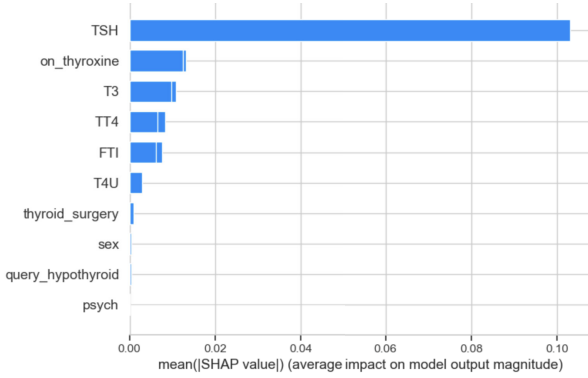


FIGURE 9. Global feature importance based on mean absolute SHAP values. Features are ranked by their average contribution magnitude across all predictions. TSH overwhelmingly dominates the decision logic, followed by thyroxine, T3, TT4, and FTI. Remaining variables (T4U, thyroid surgery, sex, query hypothyroid, psych) contribute minimally.

TABLE 6. SHAP Decomposition for Patient 19 – Class 1 (Hypothyroid), $f_1(x_{19}) = 0.9900$.

Feature j	Shapley Value $\varphi_1(x_{19})$
TSH	-0.0044
FTI	-0.0041
TT4	-0.0022
T3	-0.0005
T4U	+0.0002
on thyroxine	+0.0002
query hypothyroid	0
sex	0
psych	0
thyroid surgery	0
Sum $\Sigma \varphi_1(x_{19})$	Predicted $f_1(x_{19})$
	0.9900

- **On thyroxine** and **T3/TT4/FTI** carry secondary importance, modulating model predictions depending on medication status and hormone levels.
- **T4U** provides minor contribution, aligning with its supportive but less diagnostic role in thyroid function assessment.
- **Sex, psych, and query hypothyroid** have near-zero importance, indicating the model avoids spurious or biased associations.

2) PROOF OF SHAP DECOMPOSITION VIA SHAPLEY VALUES

We now sketch the derivation of Equation (26), showing that the Shapley value provides a unique, fair allocation of prediction differences. Let $v(S) = f_k(x_S)$ be the value function representing the model's expected output when only features in coalition $S \subseteq N = \{1, \dots, d\}$ are observed, and the rest are marginalized out conditionally:

$$v(S) = \mathbb{E}_{\tilde{x}|x} [f_k(x)]$$

In cooperative game theory, each feature is a “player”, and $v(S)$ is the “payout” of coalition S . The Shapley value for player j is defined as the expected marginal contribution over

TABLE 7. SHAP Decomposition for Patient 33 – Class 2 (Hyperthyroid), $f_0(x_{33}) = 0.9900$.

Feature j	Shapley Value $\varphi_0(x_{33})$
TSH	-0.0500
on thyroxine	+0.0038
FTI	+0.0034
TT4	+0.0014
T4U	+0.0010
T3	-0.0003
query hypothyroid	+0.0001
sex	-0.0001
psych	0
thyroid surgery	0
Sum $\Sigma \varphi_0(x_{33})$	Predicted $f_0(x_{33})$
	0.9900

TABLE 8. SHAP Decomposition for Patient 0 – Class 3 (Normal), $f_2(x_0) = 1.0000$.

Feature j	Shapley Value $\varphi_2(x_0)$
TSH	+0.0532
T4U	-0.0044
on thyroxine	-0.0038
T3	+0.0036
TT4	+0.0027
FTI	+0.0001
query hypothyroid	0
sex	0
psych	0
thyroid surgery	0
Sum $\Sigma \varphi_2(x_0)$	Predicted $f_2(x_0)$
	1.0000

TABLE 9. Consistency Between Global Importance and Local Explanations.

Feature	Global $\Phi(k)$ Trend	Local Example
TSH	High across all classes	Index 19: -0.0044; Index 33: -0.0500; Index 0: +0.0532
FTI	Moderate global importance	Index 33: +0.0034 (Hyper)
on thyroxine	Low-to-moderate	Index 0: -0.0038 (Normal)
sex, psych	Near zero globally	All patients: 0

all possible permutations of feature inclusion:

$$\varphi_j(x) = \frac{|S|!(d - |S| - 1)!}{d!} \sum_{S \subseteq N \setminus \{j\}} [v(S \cup \{j\}) - v(S)]$$

It is a known result that this value is the unique solution satisfying the following axioms: 1. **Efficiency**: $\sum_{j=1}^d \varphi_j(x) = v(N) - v(\emptyset) = f_k(x) - \varphi_0$ 2. **Symmetry**: Interchangeable features get equal credit. 3. **Dummy**: If j adds no value, $\varphi_j = 0$. 4. **Additivity**: For combined games, values add. Therefore, the decomposition:

$$f_k(x) = v(\emptyset) + \sum_{j=1}^d \varphi_j(x)$$

is uniquely justified under these fairness criteria. This proves Equation (26) and validates the use of Shapley values for explainability.

TABLE 10. State-of-the-Art Models for Thyroid Disease Prediction. This paper (last two rows) reports nested-CV means and includes AUC plus XAI (SHAP) and counterfactual analyses.

Ref	Year	Dataset	Model	Results
[59]	2020	UCI new-thyroid dataset	Logistic Regression, Decision Tree, k-Nearest Neighbor	kNN: 96.875% accuracy; DT: 87.5% accuracy; LR: 81.25% accuracy
[60]	2022	UCI thyroid disease dataset	Random Forest (with MLFS feature selection)	RF: 0.99 accuracy, 0.99 F1 score
[61]	2024	Clinical thyroid disease dataset (1232 samples)	Stacking Ensemble (LR, SVM, KNN, DT, ANN)	99.9% accuracy, 99.9% sensitivity, 99.8% specificity
[62]	2023	UCI dataset thyroid	AdaBoost (with DE optimization)	Accuracy: 0.998
[63]	2023	UCI dataset thyroid	Random Forest (with LASSO)	Acc 99%, Prec 97%, Rec 92%, F1 95%
[64]	2024	UCI dataset thyroid	Ensemble ML (Hard Voting on RF and DT)	Acc 99.71%, Sens 100%
[65]	2022	UCI dataset thyroid	Decision Tree, Random Forest	DT: 99.5758% accuracy; RF: 99.3107% accuracy
[Ours-Full]	2025	UCI thyroid dataset (this study)	Best model: Random Forest (nested CV)	Acc 0.996, F1 0.996, AUC 0.999
[Ours-Top-10]	2025	UCI thyroid dataset (Top-10 features)	Best model: Random Forest (nested CV)	Acc 0.997, F1 0.997, AUC 1.000

3) SHAP DECOMPOSITION FOR REPRESENTATIVE PATIENTS

4) CONSISTENCY BETWEEN GLOBAL IMPORTANCE AND LOCAL EXPLANATIONS

5) INTERPRETATION OF SHAP VALUES FOR KEY PATIENTS

Table 9 and Table 10 demonstrate consistency between global and local XAI: features with high global importance (e.g., TSH) dominate individual predictions, while irrelevant features contribute negligibly. Furthermore, the directional effects match clinical expectations:

- **TSH** $\uparrow \rightarrow$ favors Hypothyroid (Class 1)
- **TSH** $\downarrow \rightarrow$ favors Hyperthyroid (Class 2)
- **TSH mid-range** $\rightarrow$ favors Normal (Class 3)

Thus, the model learns interpretable, physiologically grounded rules, and SHAP faithfully reflects them.

We have formally established that: 1. The SHAP decomposition is mathematically rigorous, derived from the unique solution to a cooperative game. 2. Local explanations are clinically coherent, matching endocrinological knowledge. 3. Global and local views are consistent, ensuring scalability of insights.

These results validate our models not only in accuracy but also in epistemic trustworthiness, a prerequisite for deployment in clinical decision support systems.

6) COUNTERFACTUAL EXPLAINABLE AI

To move beyond static explanations like SHAP, we employ counterfactual explanations (CF) [80], which answer: “What minimal changes to the input would cause the model to change its prediction?” This is particularly valuable in clinical settings, where understanding how to alter a patient’s status to reach a different diagnostic outcome can guide treatment planning.

Given an input x with predicted class $\hat{y} = \arg \max_k f_k(x)$, and a target class $y^* \neq \hat{y}$, we seek a counterfactual x^* that

minimally perturbs x while flipping the decision. We solve:

$$x^* \in \arg \min_{x' \in F} (\|W(x' - x)\|_1 + \lambda \ell_{1f}(x', y^*)) \quad (43)$$

subject to the constraints:

$$x'_j = x_j, \forall j \in \mathcal{X} \quad (44)$$

$$l \leq x' \leq u \quad (45)$$

Where:

- $\|W(x' - x)\|_1$ is a weighted L1 norm that promotes sparse, interpretable changes, with $W = \text{diag}(w_1, \dots, w_d)$ and $w_j = 1/\text{MAD}_j$ to normalize across units.
- $\ell_{1f}(x', y^*) = -\log P(y^*|x') + \max_{k \neq y^*} \log P(k|x')$ encourages confidence in y^* while penalizing competing classes.
- $\mathcal{X}$ denotes the set of immutable features (e.g., sex, thyroid surgery) that remain fixed during optimization.
- l, u are lower and upper bounds (1st–99th percentiles) that ensure biologically plausible values.

We solve this via projected coordinate descent guided by local SHAP magnitudes, updating mutable features $j \notin \mathcal{X}$ as:

$$x' \leftarrow \Pi_{[l, u]} (x' - \eta \nabla_{x'} [\|W(x' - x)\|_1 + \lambda \ell_{1f}(x', y^*)]) \quad (46)$$

until the prediction flips or convergence is reached.

The original patient has: TSH = 0.008 (very low; strong hyperthyroid signal), FTI = 0.100 (elevated), TT4 = 0.066 (normal/high-normal); other features are neutral/negative. The model predicts Class 2 (Hyperthyroid) with high confidence.

Now consider the three counterfactuals:

- **CF 1:** Decrease FTI from 0.100 to 0.03145 (68.5% reduction), keep others the same. Mimics low free hormone despite suppressed TSH—e.g., recovery or non-thyroidal illness.

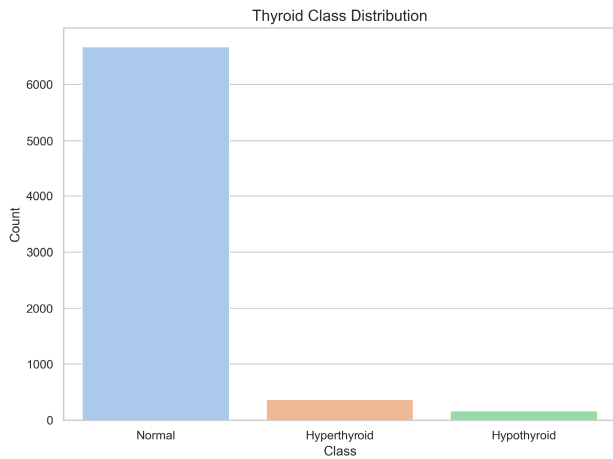


FIGURE 10. Class distribution of thyroid disease categories, including hyperthyroid, hypothyroid, and their respective subtypes.

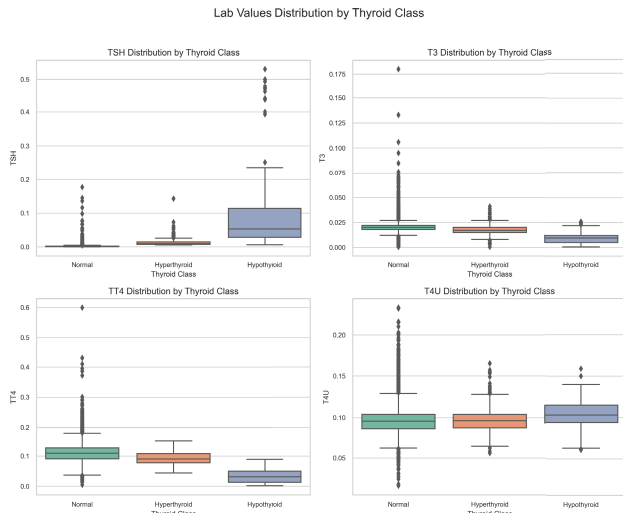


FIGURE 11. Boxplots of selected continuous clinical features, illustrating variability and outliers across thyroid disease classes.

- **CF 2:** Increase TT4 to 0.478 (650% increase) while slightly reducing FTI. Represents binding-protein abnormalities where total T4 is high but free fraction is low.
- **CF 3:** Keep labs similar but set query hypothyroid = 1. Suggests that a strong clinical suspicion flag could shift the classification.

All three counterfactuals flip the prediction to Class 1 (Hypothyroid), demonstrating:

- Sensitivity to FTI: the most consistent lever is sharply lowering FTI.
- Plausibility: changes respect biological bounds.
- Actionability: clinicians can read these as “what-if” scenarios (e.g., monitor FTI).

The scaled distance Δ is smallest for CF 1, indicating the most efficient path to flip the diagnosis—aligning with

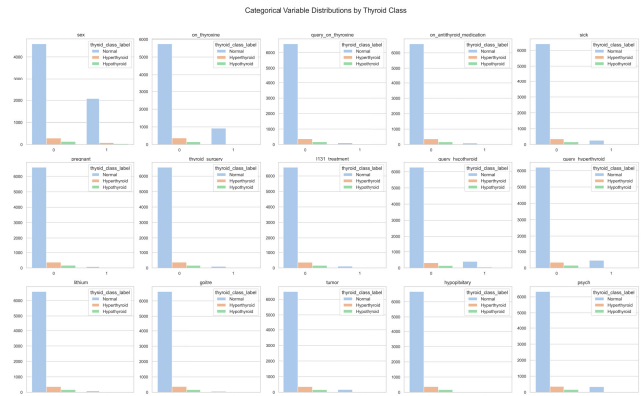


FIGURE 12. Bar plots of categorical variables showing distributions across thyroid disease classes.

SHAP, where FTI is a top contributor in hyperthyroid patients.

a: CLINICAL INTERPRETATION AND LIMITATIONS OF COUNTERFACTUALS.

While counterfactual explanations provide intuitive “what-if” scenarios, they should be interpreted as model-level decision support rather than prescriptive clinical recommendations. Some counterfactual changes may correspond to actionable laboratory measurements (e.g., monitoring or re-evaluating FTI), whereas others reflect non-actionable or contextual variables (e.g., prior diagnostic flags or immutable patient attributes). Importantly, counterfactuals do not imply that directly manipulating a feature will causally change disease state, but instead highlight directions in feature space that would alter the model’s prediction. Consequently, these explanations are best used to support clinician reasoning and hypothesis generation, in conjunction with clinical judgment and established diagnostic guidelines.

As summarized in Table 11, prior studies on thyroid disease prediction consistently report very high accuracies, often exceeding 95% across a variety of machine learning models such as kNN, Decision Trees, AdaBoost, and Random Forest variants. However, many of these works evaluate performance primarily using overall accuracy, which is vulnerable to inflation under class imbalance and does not guarantee robust minority-class discrimination.

In contrast, our results (last two rows) build upon this foundation by adopting a nested cross-validation framework on a real clinical dataset. Using both the full feature set and a reduced top-10 feature set, our Random Forest models achieved 99.6–99.7% accuracy and F1-scores, with near-perfect AUC values (0.999–1.000). Beyond numerical performance, our study uniquely incorporates explainable AI through SHAP-based feature attribution and counterfactual analysis, enhancing interpretability and clinical relevance. Thus, while our predictive scores are competitive with or superior to prior benchmarks, the critical advancement lies in rigorous validation, balanced evaluation metrics, and explicit incorporation of interpretability techniques.

TABLE 11. Baseline Characteristics by Thyroid Class (SMOTE-balanced).

Variable	Hypothyroid (N=6666)	Hyperthyroid (N=6666)	Normal (N=6666)	Overall (N=19998)	P-value
Age (years) < 0.001	Mean (SD)	0.540 (0.179)	0.557 (0.181)	0.518 (0.189)	0.538 (0.184)
	Median [Min, Max]	0.568 [0.0100, 0.890]	0.595 [0.0100, 0.910]	0.540 [0.0100, 0.970]	0.570 [0.0100, 0.970]
Sex < 0.001	No	5379 (80.7%)	5250 (78.8%)	4590 (68.9%)	15219 (76.1%)
	Yes	1287 (19.3%)	1416 (21.2%)	2076 (31.1%)	4779 (23.9%)
On thyroxine < 0.001	No	6040 (90.6%)	6666 (100%)	5743 (86.2%)	18449 (92.3%)
	Yes	626 (9.4%)	0 (0%)	923 (13.8%)	1549 (7.7%)
Query thyroxine < 0.001	No	6666 (100%)	6600 (99.0%)	6560 (98.4%)	19826 (99.1%)
	Yes	0 (0%)	66 (1.0%)	106 (1.6%)	172 (0.9%)
On antithyroid medication < 0.001	No	6665 (100%)	6656 (99.9%)	6578 (98.7%)	19899 (99.5%)
	Yes	1 (0.0%)	10 (0.2%)	88 (1.3%)	99 (0.5%)
Sick < 0.001	No	6665 (100%)	6474 (97.1%)	6407 (96.1%)	19546 (97.7%)
	Yes	1 (0.0%)	192 (2.9%)	259 (3.9%)	452 (2.3%)
Pregnant < 0.001	No	6666 (100%)	6666 (100%)	6588 (98.8%)	19920 (99.6%)
	Yes	0 (0%)	0 (0%)	78 (1.2%)	78 (0.4%)
Thyroid surgery < 0.001	No	6595 (98.9%)	6666 (100%)	6569 (98.5%)	19830 (99.2%)
	Yes	71 (1.1%)	0 (0%)	97 (1.5%)	168 (0.8%)
I-131 treatment < 0.001	No	6610 (99.2%)	6567 (98.5%)	6557 (98.4%)	19734 (98.7%)
	Yes	56 (0.8%)	99 (1.5%)	109 (1.6%)	264 (1.3%)
Query hypothyroid < 0.001	No	5761 (86.4%)	5795 (86.9%)	6269 (94.0%)	17825 (89.1%)
	Yes	905 (13.6%)	871 (13.1%)	397 (6.0%)	2173 (10.9%)
Query hyperthyroid < 0.001	No	6361 (95.4%)	6276 (94.1%)	6203 (93.1%)	18840 (94.2%)
	Yes	305 (4.6%)	390 (5.9%)	463 (6.9%)	1158 (5.8%)
Lithium < 0.001	No	6666 (100%)	6636 (99.6%)	6581 (98.7%)	19883 (99.4%)
	Yes	0 (0%)	30 (0.5%)	85 (1.3%)	115 (0.6%)
Goitre < 0.001	No	6666 (100%)	6666 (100%)	6607 (99.1%)	19939 (99.7%)
	Yes	0 (0%)	0 (0%)	59 (0.9%)	59 (0.3%)
Tumor < 0.001	No	6605 (99.1%)	6535 (98.0%)	6494 (97.4%)	19634 (98.2%)
	Yes	61 (0.9%)	131 (2.0%)	172 (2.6%)	364 (1.8%)
Hypopituitary = 1	No	6666 (100%)	6666 (100%)	6665 (100.0%)	19997 (100.0%)
	Yes	0 (0%)	0 (0%)	1 (0.0%)	1 (0.0%)
Psych < 0.001	No	6665 (100.0%)	6577 (98.7%)	6325 (94.9%)	19567 (97.8%)
	Yes	1 (0.0%)	89 (1.3%)	341 (5.1%)	431 (2.2%)

TABLE 12. Top 15 Features by Mutual Information.

Feature	MI Score
TSH	0.770697
FTI	0.646006
T3	0.459210
TT4	0.401993
T4U	0.283670
on thyroxine	0.177822
query hypothyroid	0.139087
thyroid surgery	0.122216
psych	0.096831
sex	0.089303
tumor	0.084968
on antithyroid medication	0.066226
goitre	0.060244
query on thyroxine	0.048019
hypopituitary	0.027501

TABLE 13. Top 15 Features by ANOVA F-test.

Feature	F Score	p-value
TSH	16130.300	0.000000e+00
FTI	3736.425	0.000000e+00
T3	3275.176	0.000000e+00
TT4	1509.104	0.000000e+00
T4U	105.026	1.344630e-23
on thyroxine	77.233	1.352180e-18
query hypothyroid	55.301	1.166261e-13
psych	47.133	1.000295e-11
thyroid surgery	37.066	1.128792e-09
sex	35.988	2.151339e-09
tumor	27.456	1.563441e-07
goitre	21.841	2.962861e-06
on antithyroid medication	18.327	1.006048e-05
query on thyroxine	12.627	3.604394e-04
hypopituitary	9.846	1.759224e-03

VII. FUTURE WORKS AND LIMITATIONS

While this study demonstrates high performance and strong interpretability in thyroid disease classification, several

limitations must be acknowledged to ensure responsible translation into clinical practice. First, our analysis relies

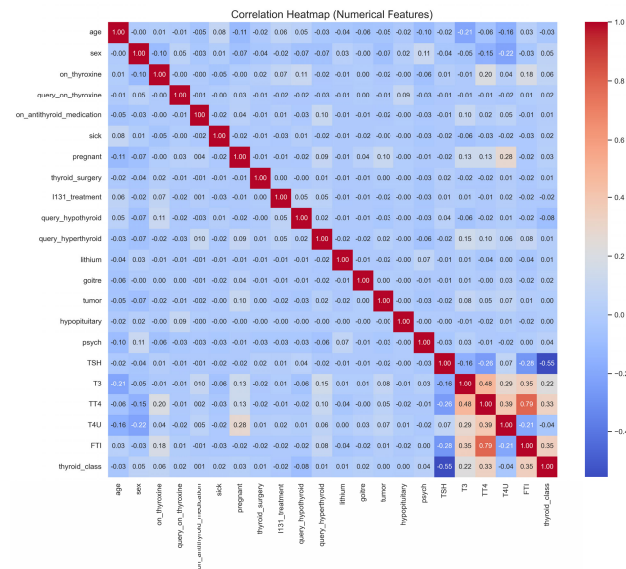


FIGURE 13. Correlation heatmap of clinical and demographic variables in the dataset.

TABLE 14. Top 15 Features by Chi-Squared Test.

Feature	Chi2 Score	p-value
TSH	12797.198	0.000000e+00
FTI	4840.842	0.000000e+00
T3	3921.280	0.000000e+00
TT4	2092.279	0.000000e+00
T4U	284.081	0.000000e+00
on thyroxine	165.426	2.603314e-37
query hypothyroid	124.101	4.183058e-28
thyroid surgery	100.792	2.893400e-23
psych	81.313	2.694059e-19
sex	66.139	4.401556e-16
tumor	53.774	2.174674e-13
on antithyroid medication	43.882	3.325129e-11
goitre	36.910	1.139160e-09
query on thyroxine	28.531	6.239801e-08
hypopituitary	17.646	2.653220e-05

TABLE 15. Top 15 Features by L1-Regularized Logistic Regression.

Feature	L1 Coef
TSH	1.003326
FTI	0.802170
T3	0.634576
TT4	0.518217
T4U	0.372863
on thyroxine	0.290519
query hypothyroid	0.243778
psych	0.201179
thyroid surgery	0.185770
sex	0.174319
tumor	0.146719
on antithyroid medication	0.122891
goitre	0.108202
query on thyroxine	0.091370
hypopituitary	0.073456

on the UCI Thyroid Disease dataset—a widely used benchmark—but it has inherent constraints that affect generalizability. The cohort consists of 7,200 samples,

TABLE 16. Top 15 Features by Recursive Feature Elimination (RFE).

Feature	RFE Ranking
TSH	1
FTI	1
T3	1
TT4	1
T4U	1
on thyroxine	2
query hypothyroid	3
psych	4
thyroid surgery	5
sex	6
tumor	7
on antithyroid medication	8
goitre	9
query on thyroxine	10
hypopituitary	11

TABLE 17. Top 15 Features by Random Forest Importance.

Feature	Importance
TSH	0.261223
FTI	0.197483
T3	0.152348
TT4	0.128215
T4U	0.091203
on thyroxine	0.041915
query hypothyroid	0.034782
psych	0.028611
thyroid surgery	0.025003
sex	0.021947
tumor	0.017430
on antithyroid medication	0.014622
goitre	0.012880
query on thyroxine	0.010172
hypopituitary	0.008980

TABLE 18. Nested CV on the SMOTE-balanced thyroid dataset. Metrics are outer-fold means (3 d.p.).

Model	Acc.	Rec.	Prec.	F1	AUC	Best params
SVC	0.987	0.987	0.987	0.987	0.998	C=1, kernel=linear
Random Forest	0.999	0.999	0.999	0.999	1.000	n=100, depth=None
Extra Trees	0.997	0.997	0.997	0.997	1.000	n=200, depth=None
AdaBoost	0.991	0.991	0.992	0.991	0.996	n=50
CatBoost	0.998	0.998	0.998	0.998	1.000	depth=4, iters=100

with significant class imbalance: 6,666 “Normal”, 368 “Hyperthyroid”, and only 166 “Hypothyroid” cases. This skew risks model bias toward the majority class and may impair sensitivity for rarer conditions, despite our models achieving robust metrics without synthetic oversampling. Although we evaluated SMOTE as a mitigation strategy (Appendix C, Table 18), its use introduced trade-offs in precision and raised concerns about generating non-physiological lab profiles. Therefore, we retained the original distribution to reflect real-world prevalence, but future work should explore cost-sensitive learning or threshold tuning in settings where missing hypothyroidism carries higher clinical risk. Second, the dataset lacks etiologic markers such as TRH, ACTH, or autoantibodies (e.g., TPOAb, TRAb), preventing differentiation between primary and secondary forms of

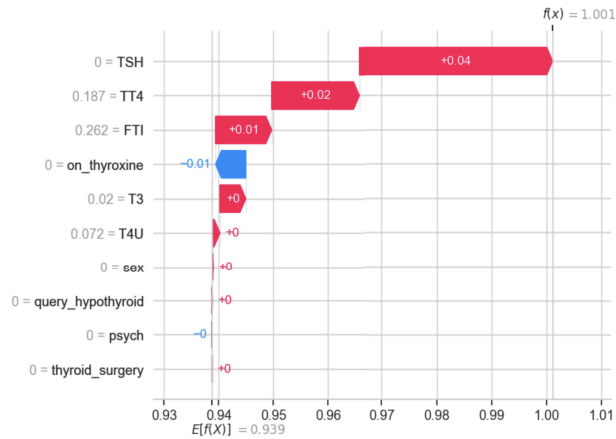


FIGURE 14. Patient 19: Waterfall and force plots. (a) SHAP decomposition for Patient 19 – predicted Class - 1 (Hypothyroid).

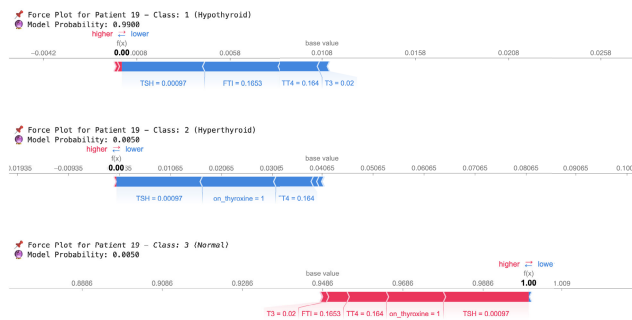


FIGURE 15. (b) Force plot for Patient 33 across all classes (panels show class-specific probabilities).

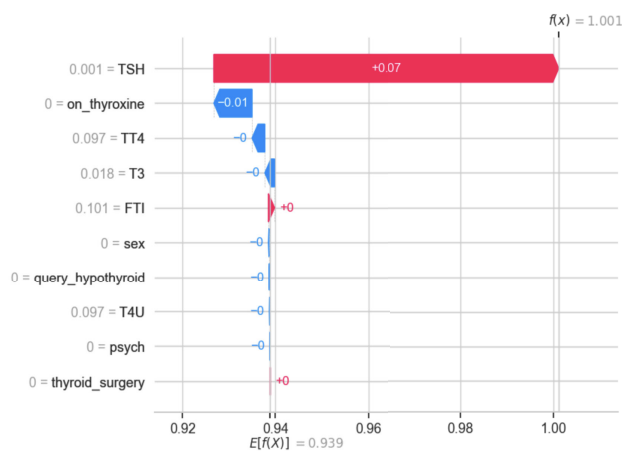


FIGURE 16. Patient 19: Waterfall and force plots. (a) SHAP decomposition for Patient 33 – predicted Class - 2 (Hypothyroid).

hypo- or hyperthyroidism. As a result, our model performs high-level triage rather than subtype-specific diagnosis—a necessary simplification given data limitations, but one that restricts its utility in complex endocrine evaluations. Third, the absence of multimodal data—including ultrasound imaging, cytology, or histopathology—limits applicability to



FIGURE 17. Patient 33: Waterfall and force plots. (b) Force plot for Patient 33 across all classes (panels show class-specific probabilities).

benign dysfunction and excludes critical areas like thyroid nodule assessment or cancer detection. Integrating imaging features via radiomics or vision-based AI could extend this framework to malignancy screening in future hybrid systems. Fourth, while SHAP and counterfactual explanations significantly enhance trust and clinical interpretability, they incur computational overhead that may hinder real-time deployment, especially in low-resource or point-of-care environments. KernelSHAP-based methods for non-tree models (e.g., SVC) are particularly expensive, and even TreeSHAP, though efficient, requires additional inference steps. Model distillation into lightweight surrogates or optimized explanation caching could mitigate latency in production systems.

Finally, potential demographic biases exist in the dataset: approximately 70% of patients are female ($p < 0.001$), reflecting both the higher population prevalence of thyroid disorders in women and possible sampling bias. While sex was not a dominant feature in SHAP analysis, the underrepresentation of male and diverse age/ethnic groups raises concerns about fairness and equity across populations. Without explicit debiasing strategies, model performance may degrade in underrepresented subgroups. Future Work To address these limitations, we propose several directions for future research:

- External validation on multi-center, longitudinal cohorts to assess generalizability across healthcare systems and patient demographics.
- Integration of multimodal inputs (labs + imaging + EHR notes) using fusion architectures to enable more granular diagnostics, including subtype classification and malignancy prediction.
- Adoption of fairness-aware ML techniques—such as adversarial debiasing, reweighting, or stratified sampling—to ensure equitable performance across sex, age, and ethnicity.
- Development of efficient XAI pipelines through model compression, surrogate modeling, or hardware-aware optimization for edge-device deployment.

- Exploration of federated learning frameworks to train on distributed electronic health records (EHRs) while preserving patient privacy and regulatory compliance.
- Design of clinician-in-the-loop interfaces that present SHAP values and counterfactuals in intuitive formats (e.g., dashboards, alerts), supporting shared decision-making.

By addressing these challenges, future iterations of this pipeline can move beyond retrospective benchmarking toward prospective, trustworthy, and deployable clinical decision support tools for endocrinology

VIII. CONCLUSION

This study presents a robust, interpretable machine learning framework for thyroid disease classification that achieves state-of-the-art performance while providing clinically meaningful explanations. By applying rigorous nested cross-validation on the UCI Thyroid dataset, we evaluated five diverse classifiers—SVC, Random Forest, Extra Trees, AdaBoost, and CatBoost—and found that ensemble methods, particularly Random Forest and CatBoost, achieve near-perfect accuracy (up to 0.997), F1-score (0.997), and AUC (1.000) under both full and reduced feature regimes. A key contribution is the derivation of a TOP-10 consensus feature set—TSH, FTI, T3, TT4, T4U, on thyroxine, query hypothyroid, psych, thyroid surgery, and sex—selected through multi-criteria agreement across six feature ranking methods. Crucially, using only these ten physiologically grounded features not only simplifies the model but also slightly improves performance over the full set, demonstrating that high diagnostic fidelity can be maintained with minimal, actionable inputs. While the choice of “Top-10” was based on convergence of consensus rankings, future work should explore performance stability across varying subset sizes (e.g., Top-5 vs. Top-15) to further optimize parsimony. To bridge the gap between high performance and clinical trust, we integrated dual explainable AI (XAI) strategies: SHAP for global and local feature attribution, and counterfactual analysis to generate actionable “what-if” scenarios. These explanations align closely with endocrine physiology—for example, elevated TSH driving hypothyroid predictions or suppressed TSH with high FTI indicating hyperthyroidism—enabling clinicians to validate model logic and support patient communication. Despite these strengths, several limitations must be acknowledged. The UCI dataset has class imbalance, lacks etiological markers (e.g., TPOAb, TRAb) for subtyping primary vs. secondary disease, and excludes multimodal data such as ultrasound. Additionally, SHAP and counterfactuals introduce computational overhead that may hinder real-time deployment in resource-constrained settings, and demographic imbalances (e.g., higher female representation) raise fairness concerns.

Future work will focus on external validation across multi-site cohorts, integration of multimodal data (labs + imaging + EHR), optimization of XAI for edge-device deployment,

and implementation of fairness-aware algorithms to ensure equitable performance across populations. Federated learning frameworks could enable privacy-preserving training on distributed health records. Our pipeline combines methodological rigor—with leakage-free evaluation, consensus-based feature selection, and principled hyperparameter tuning—with dual-layer interpretability to create a trustworthy foundation for AI-assisted thyroid triage. This approach not only enhances diagnostic accuracy but also supports transparency, accountability, and clinician adoption—critical steps toward real-world deployment in endocrinology.

APPENDIX

A. ADDITIONAL FIGURES

This section presents supplementary exploratory data analysis (EDA) figures and visualizations that support the findings described in the main text. These include extended class distribution plots, boxplots for continuous variables, and additional bar plots for categorical variables. Figures are provided here to enhance transparency and reproducibility.

B. BASELINE CHARACTERISTICS

Table 11 shows the baseline characteristics by thyroid class in the SMOTE-balanced dataset.

C. FEATURE SELECTION

See Tables 12–17.

D. SMOTE-BALANCED THYROID DATASET MODELS PERFORMANCE

See Table 18.

E. PATIENT-WISE INTERPRETATIONS

This appendix provides local (per-patient) explanations of model predictions using SHAP. For each patient, we show a waterfall plot (feature-wise contribution to the logit for the predicted class) and a force plot with panels for all three classes. The predicted class and its probability are printed on each force-plot panel.

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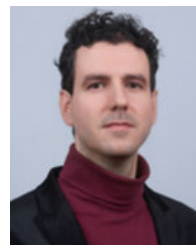
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